

PARTS CATALOGUE
VOLTAS ELECTRIC FORKLIFT
MODEL : EVX 20 MAX
(Common for all 2T AC/DC & Fully AC)



PART NO :

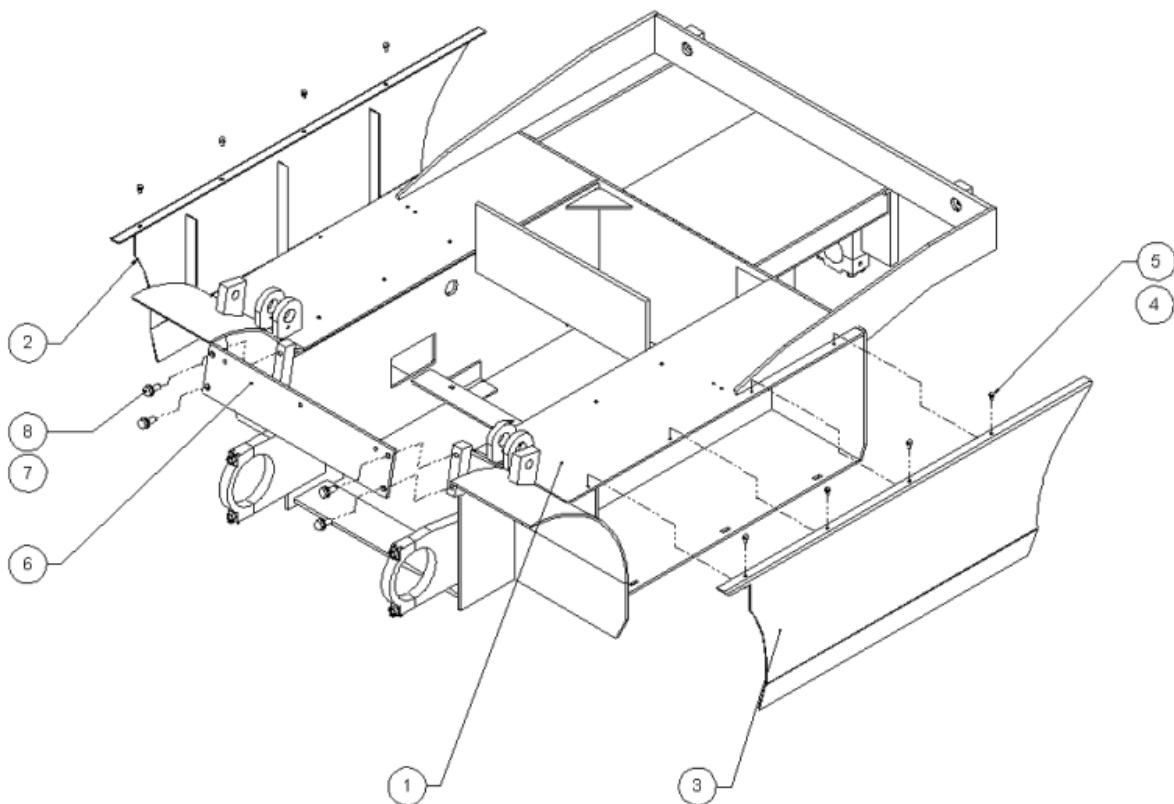
EVX 15-20 MAX

VALID FOR TRUCK

MAY 2017



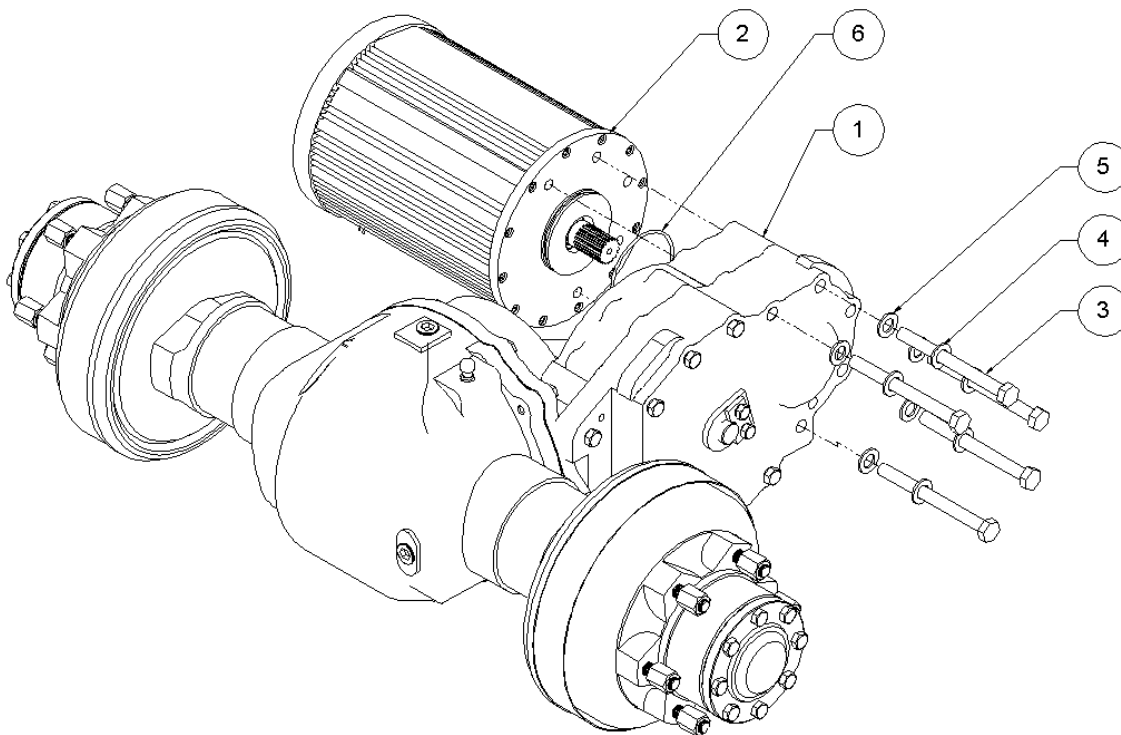
FRAME ASSEMBLY



FRAME ASSEMBLY

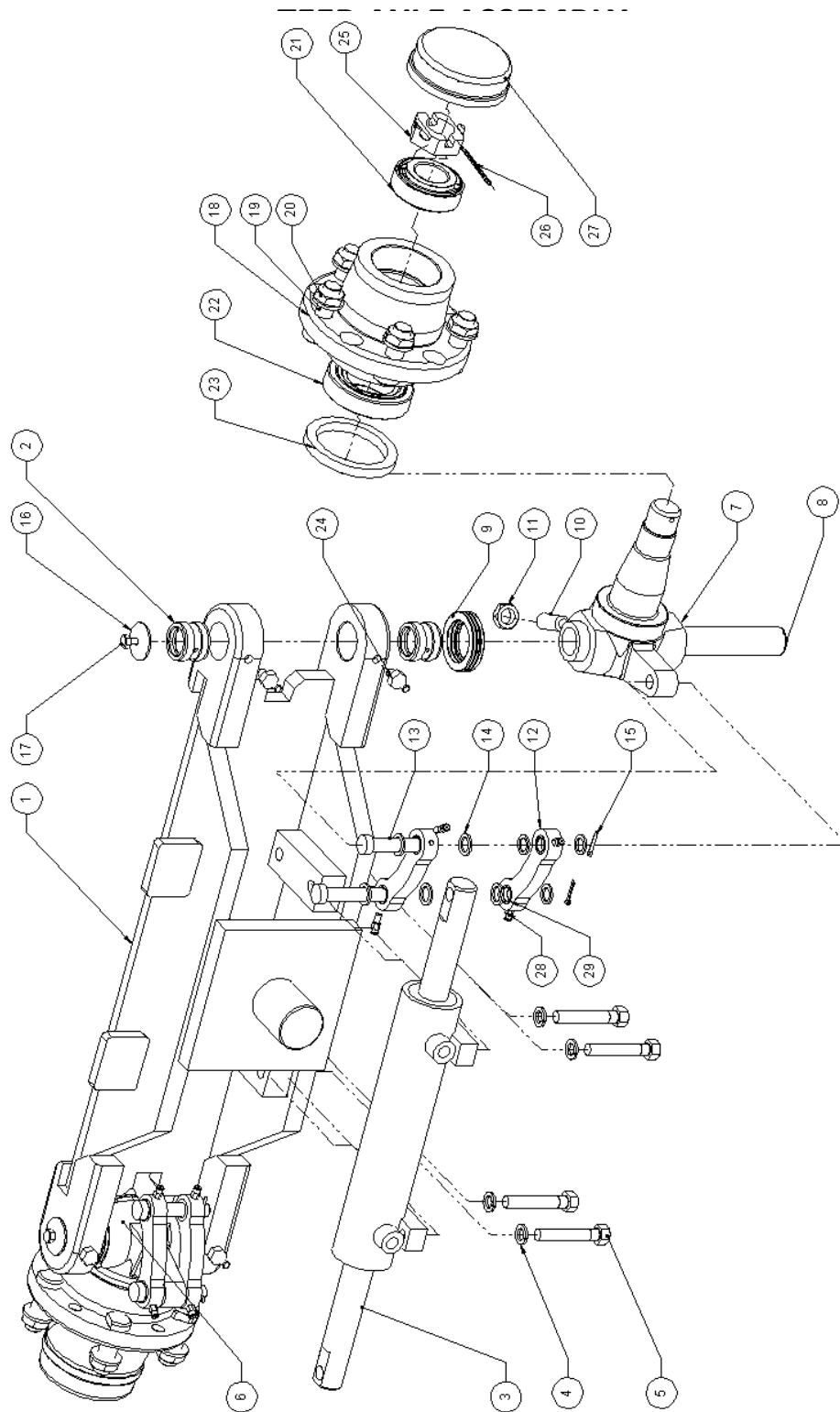
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
1	MH799201121	FRAME WELD'T	1	
2	MH799215039	SIDE COVER RIGHT SIDE	1	
3	MH799215023	SIDE COVER LEFT SIDE	1	
4	MH992791515	SPRING WASHER	8	M6
5	MH992275212	HEX. HEAD BOLT	8	M6x12LG.
6	MH799201131-06	THICK PLATE	1	562x120x8mm
7	MH992785644	PLAIN WASHER	4	M12
8	MH992276377	HEX. HEAD BOLT	4	M6x30LG.
	MH796412512	Seat Assy	1	
	MH799215042	FRP Battery Top Cover	1	
	MH16940	Seat Belt	1	

DRIVE AXLE ASSEMBLY



DRIVE AXLE ASSEMBLY

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
1	MH40497	Gear Box + AXLE	1	
1A	MH40495	GEAR BOX	1	
1B	MH40496	DRIVE AXLE	1	
1C		GASKET	1	
2	*MH40493	DRIVE MOTOR(CFR)	1	
3	MH992186484	HEX. HEAD BOLT	5	M14x155LG.
4	MH992791555	SPRING WASHER	5	M14
5	MH992781654	PLAIN WASHER	5	M14
6	MH015492400	O-RING	1	
	*MH40942	AC Traction Motor (PM)	1	
	MH42480	Encoder Hall Effect	1	
	MHENCHE469	Encoder Probe Type	1	64 Pulse

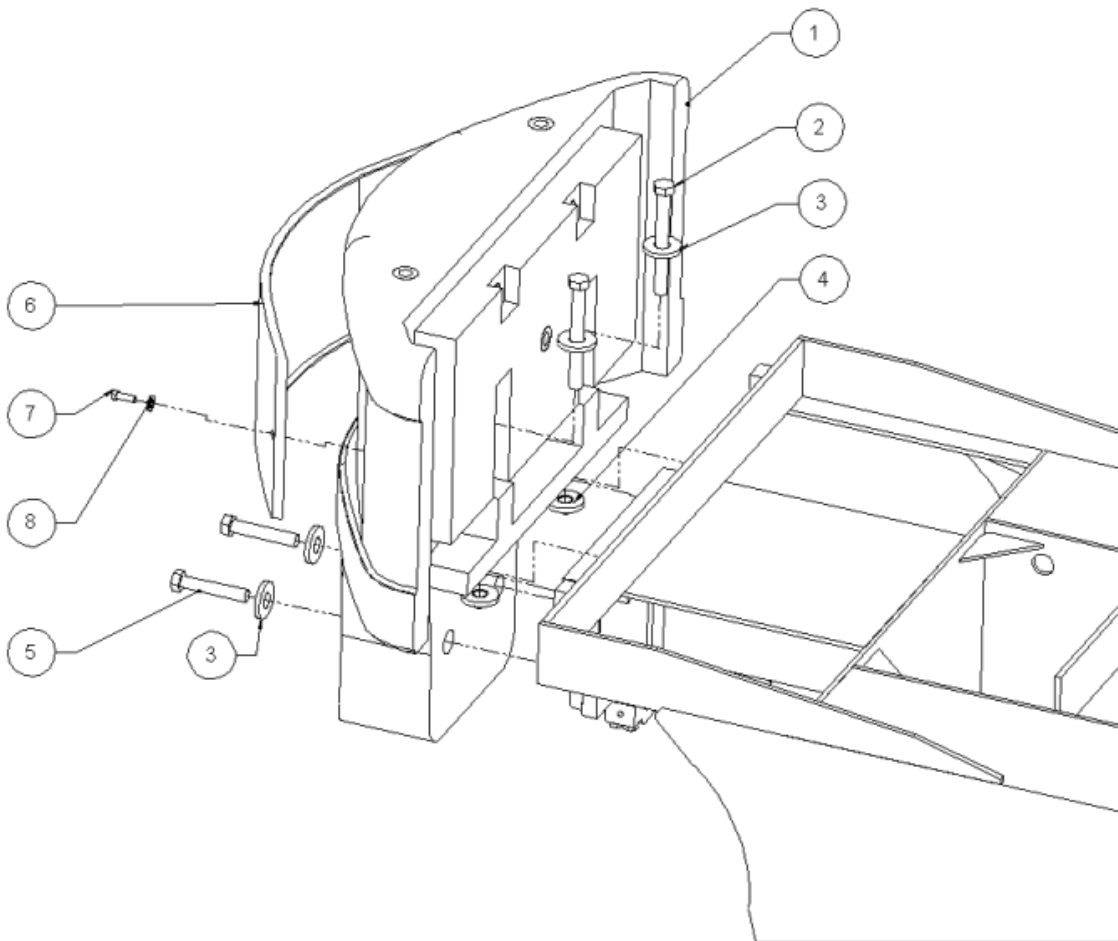


ITEM NO.	PART NUMBER	DESCRIPTION	QTY	REMARKS
1	MH799203081	STEER AXLE WELD'T FINISHED	1	
2	MH798207133	BUSH FOR STEERING KNUCKLE	4	
3	*MH42494	STEER CYLINDER Sundaram (Sim MH42033)	1	
4	MH992791545	SPRING WASHER M12	4	
5	MH992175427	HEX. HEAD BOLT	4	M12x70 LG.
6	MH799203102	STEER KNUCKLE L.H	1	
7	MH799203092	STEER KNUCKLE R.H.	1	
8	MH798207143	STEERING KNUCKLE PIN	2	
9	MH798207174	THRUST BEARING	2	
10	MH992573501	GRUB SCREW	2	M16x40LG.
11	MH992763017	LOCK NUT	2	M16x1.5P
12	MH799203114	CONNECTING LINK ASSEMBLY	4	
13	MH17265	HEADED PIN	4	
14	MH992785654	PLAIN WASHER	16	M14
15	MH992826220	SPLIT PIN	4	Ø3x20LG.
16	MH23818300	EXPANSION PLUG	2	
17	MH992171216	HEX. HEAD BOLT	2	M6x20 LG.

18	MH7982071 22	REAR WHEEL HUB	2	
19	MH7982071 54	TRAIL WHEEL BOLT	10	
20	MH7982071 64	TRAIL WHEEL NUT	10	
21	MH9933611 37	OUTER TAPER ROLLER BEARING	2	
22	MH9933611 40	INNER TAPER ROLLER BEARING	2	
23	MH9934321 35	OIL SEAL	2	
24	MH7853243 00	GREASE NIPPLE	4	
MH797903013		Bushing trail axle	2	
MH799209448		Hose Kit Stear Axle	1	

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
25	MH014984900	SLOTTED NUT	2	1 1/4" UNF
26	MH015420400	SPLIT PIN	2	
27	MH514008300	HUB CAP	2	
28	MH993791106	GREASE NIPPLE	8	M6
29	MH09186	BUSH	8	

COUNTERWEIGHT ASSEMBLY

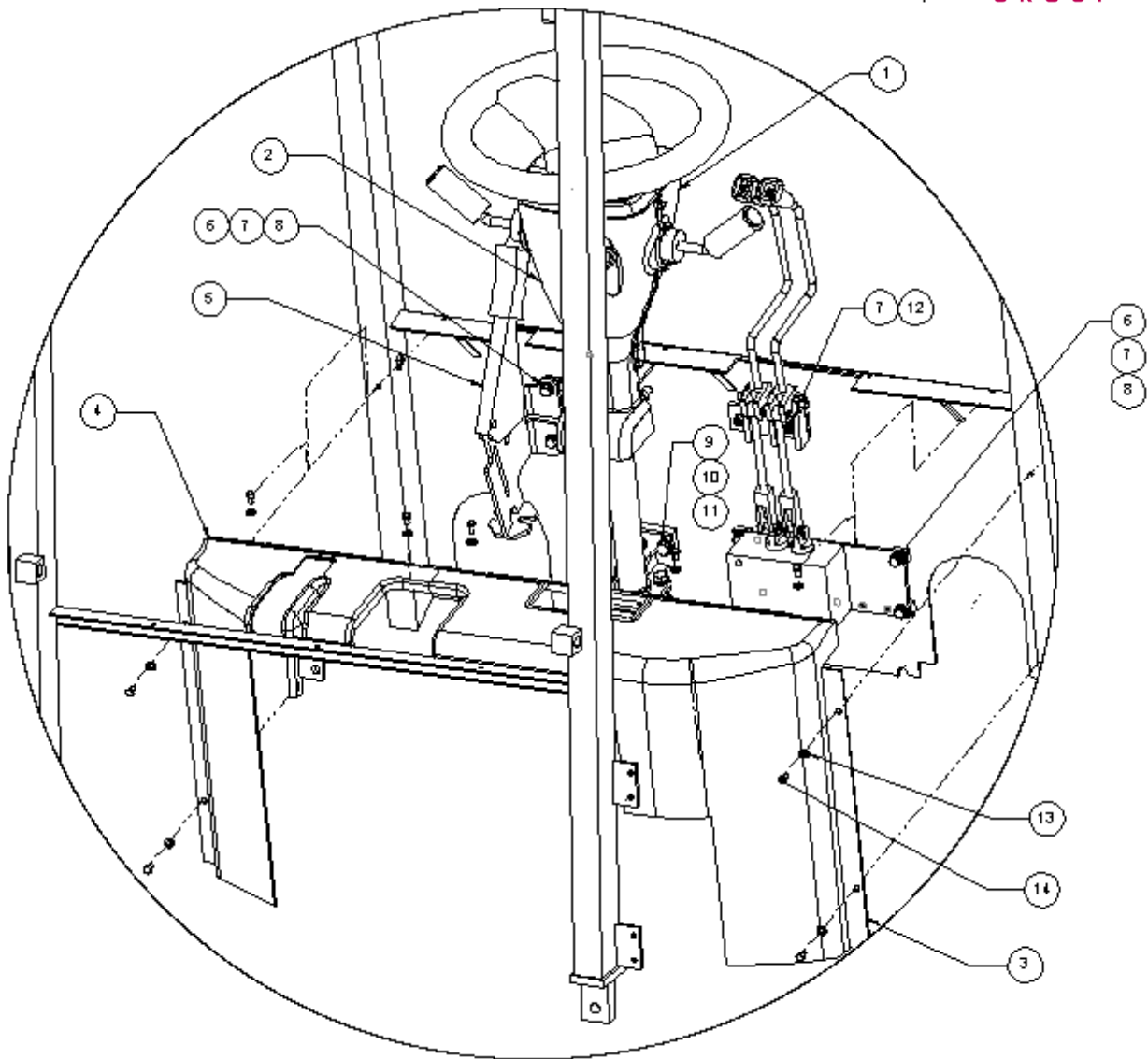


COUNTERWEIGHT ASSEMBLY

ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
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1	MH799204031	COUNTERWEIGHT ASSEMBLY	1	
2	MH992176621	HEX. HEAD BOLT	2	M20x180 LG.
3	MH870001204	SPECIAL WASHER	4	OD60XID26X6
4	MH08724	HEX. NUT	2	M20
5	MH992176614	HEX. HEAD BOLT	2	M20x110 LG.
6	MH799204043	COUNTERWEIGHT COVER	1	
7	MH992271421	HEX. HEAD BOLT	2	M12x40LG.
8	MH992785644	PLAIN WASHER	2	M12

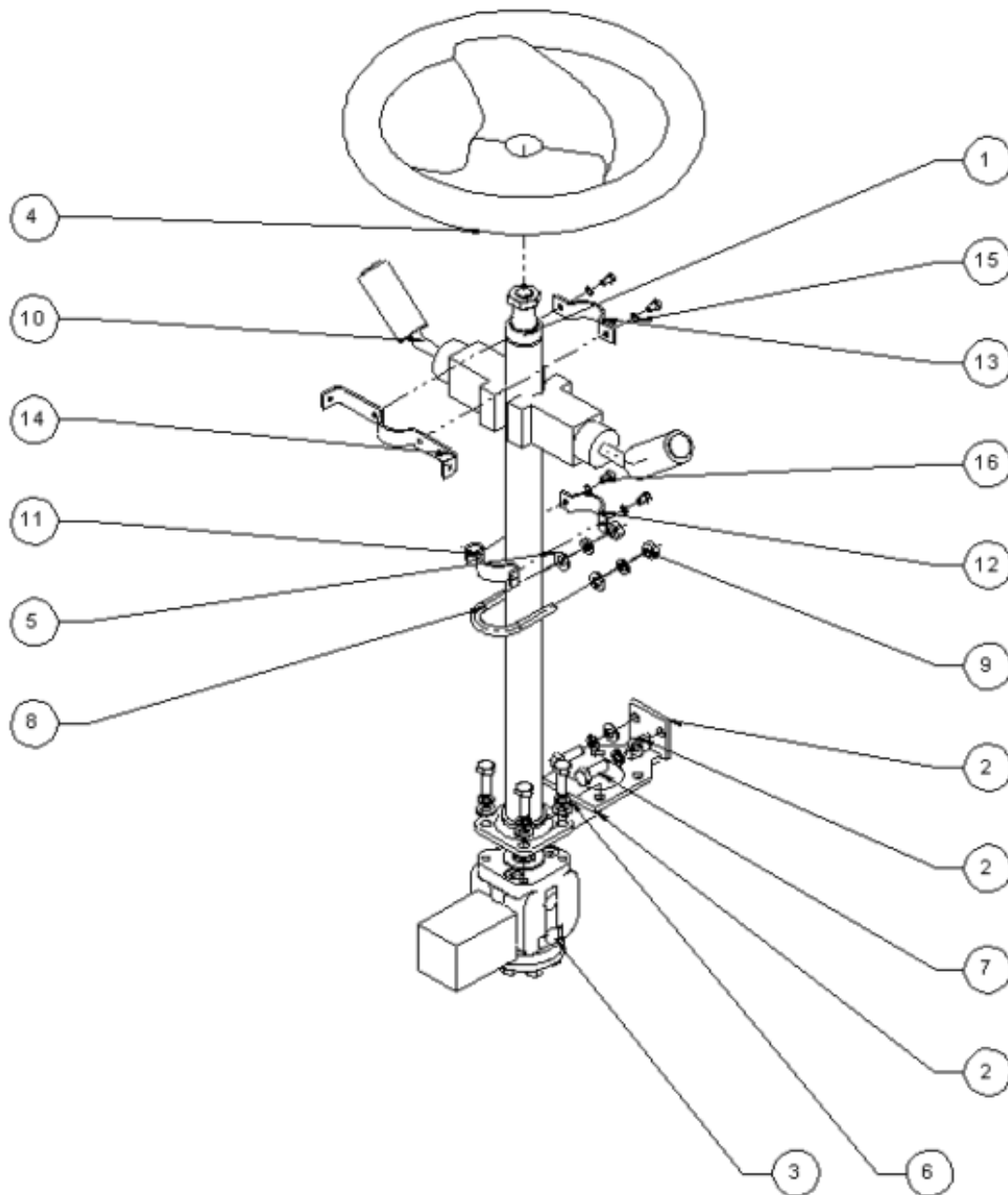
COWL ASSEMBLY



COWL ASSEMBLY

ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH799215063	STEERING COVER BACK	1	
2	MH799215053	STEERING COVER FRONT	1	
3	MH797905233	PARKING BRAKE	1	
4	MH799215082	LH COWL COVER	1	
5	MH799215012	RH COWL COVER	1	
6	MH992275295	HEX. HEAD BOLT	6	M8x20 LG.
7	MH992791525	SPRING WASHER	8	M8
8	MH992781624	PLAIN WASHER	6	M8
9	MH992275377	HEX. HEAD BOLT	2	M10x30 LG.
10	MH992791535	SPRING WASHER	2	M10
11	MH992781634	PLAIN WASHER	2	M10
12	MH992571295	HEX. SOCKET HEAD CAP SCREW	2	M8x20 LG.
13	MH992781604	PLAIN WASHER	9	M6
14	MH992275215	HEX. HEAD BOLT	9	M6x20 LG.

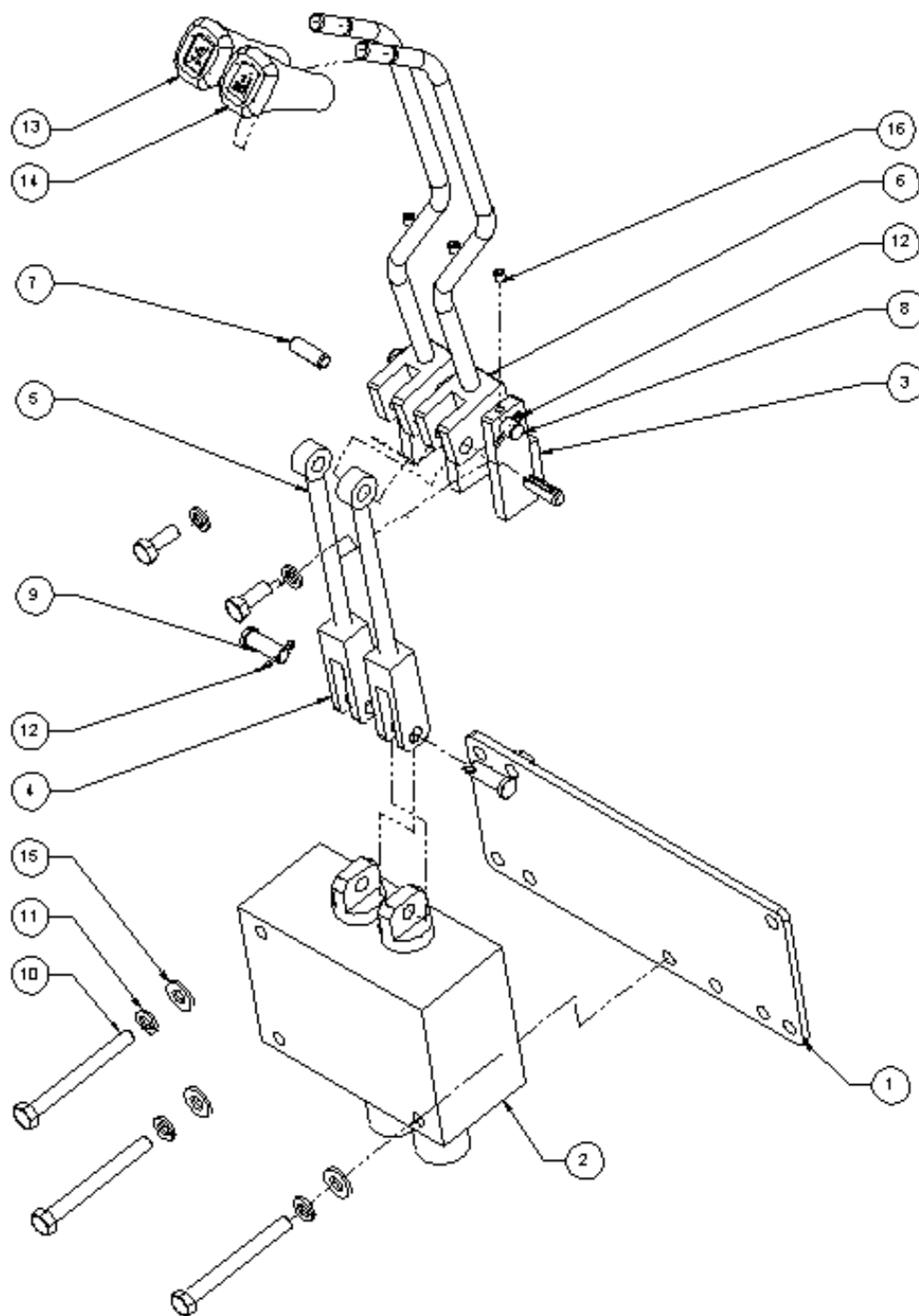
STEERING COLUMN ASSEMBLY



STEERING COLUMN ASSEMBLY

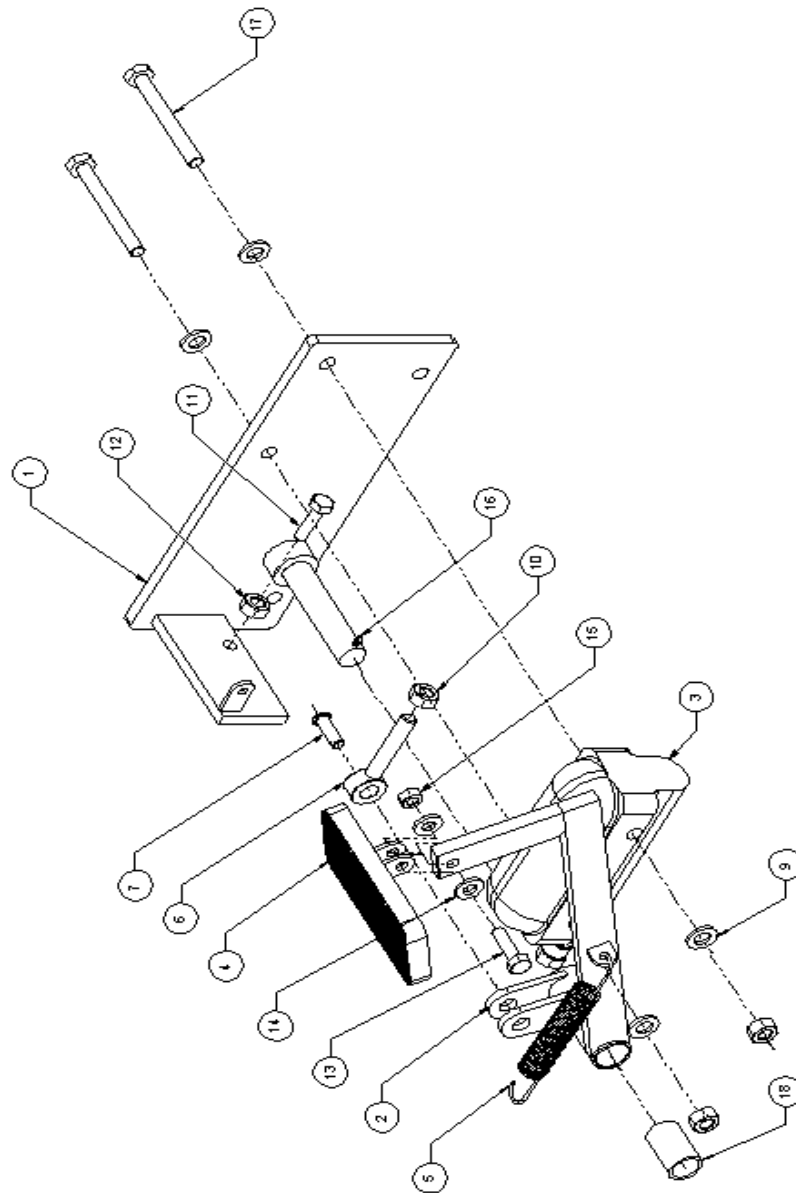
ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH799804023	STEERING COLUMN	1	
2	MH799213043	STEERING COLMN. MTG. BKT.	1	
3	MH42046	STEERING UNIT	1	
4	*MH796415029	STEERING WHEEL	1	
5	MH992781634	PLAIN WASHER	8	M10
6	MH992791535	SPRING WASHER	8	M10
7	MH992275377	HEX. HEAD BOLT	6	M10x30 LG.
8	MH13920	U' BOLT	1	
9	MH992761214	HEX. NUT	2	M10
10	MH40468	COMBINATION SWITCH	1	
11	MH799215104	SMALL STEERING BRKT FRONT	1	
12	MH799215114	SMALL STEERING BRACKET CLAMP	1	
13	MH799215134	STEERING BRACKET CLAMP	1	
14	MH799215124	STEERING BRACKET FRONT	1	
15	MH992791515	SPRING WASHER	4	M6
16	MH992571214	HEX. HEAD BOLT	6	M6x16 LG.
	*MH42109	Steering Wheel(Dia 325)	1	
	MH799213034	Cowl Mounting Brkt	1	

CONTROL VALVE LINKAGE ASSEMBLY



ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH799209213	CONTROL VALVE PLATE	1	
2	MH42052	2-SPOOL CONTROL VALVE WALVOIL	1	
3	MH799209224	LEVER BKT WELD'T	1	
4	MH799209264	YOKE END	2	
5	MH799209254	AUX CONTROL ROD	2	
6	MH799209233	HOIST HANDLE WELD'T	2	
7	MH500067906	ROLL PINS	2	
8	MH799209244	Handle PIN	1	
9	MH799209274	Yoke PIN	2	
10	MH992275295	HEX. HEAD BOLT	4	M8x80 LG.
11	MH992791525	SPRING WASHER	3	M8
12	MH992826140	SPLIT PIN	4	Ø3 X 20LG.
13	MH798722214	KNOB - CONTROL VALVE TILT	1	
14	MH798722224	KNOB - CONTROL VALVE HOIST	1	
15	MH992781624	PLAIN WASHER	3	M8
16	MH992573176	GRUB SCREW	3	M5 X 6LG.

MH12969
LIMIT S/W
2



BRAKE ASSEMBLY

EVX 15-20 MAX

MAY 2017

BRAKE ASSEMBLY

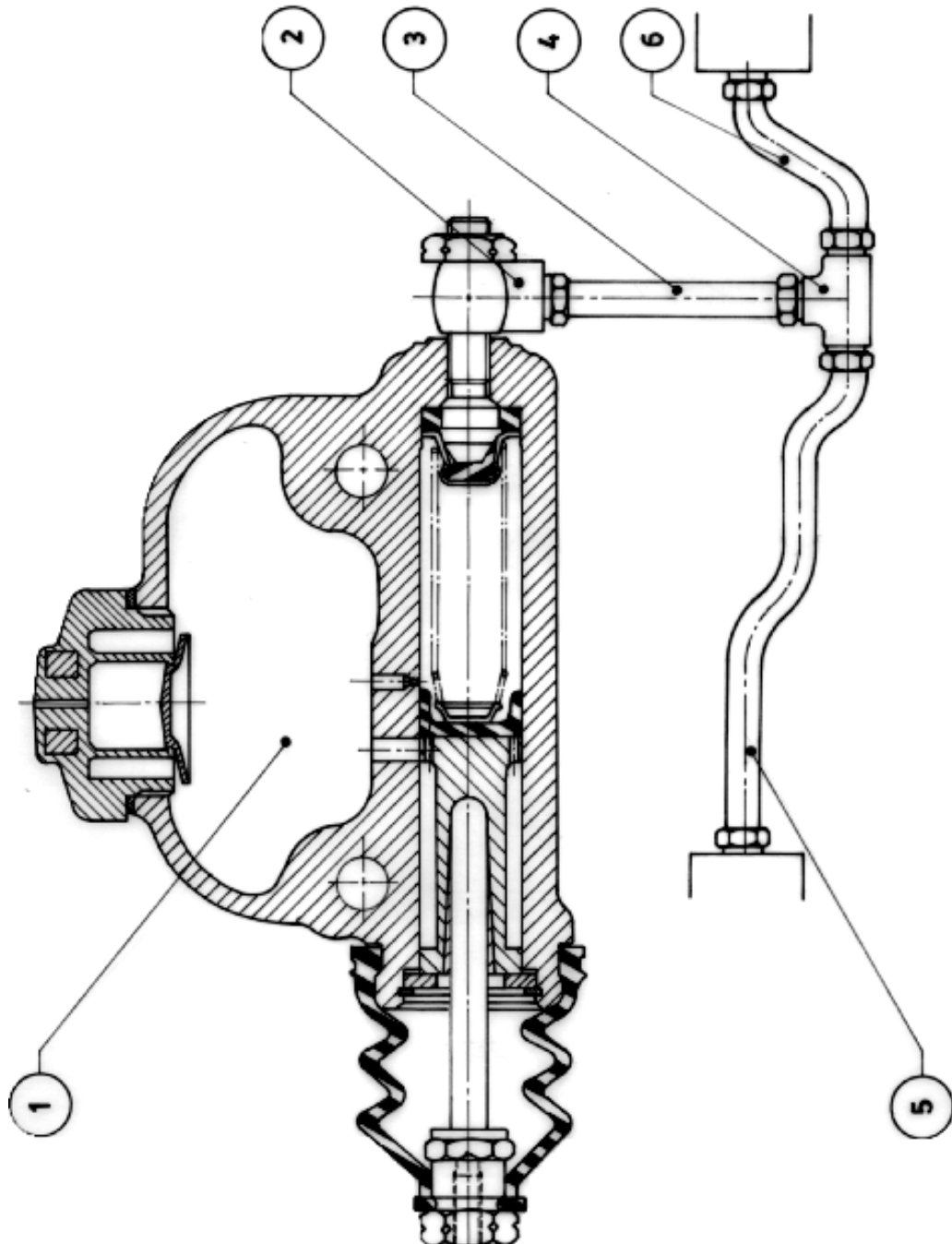
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
1	MH799214023	BRAKE MOUNTING PLATE	1	
2	MH799214033	BRAKE LEVER WELD'T	1	
3	MH870001211	MAST CYL ASSY	1	7/8"
4	MH063681200	PEDAL PAD	1	
5	MH60309600	RETURN SPRING	1	
6	MH799214064	ROD END	1	
7	*MH785577603	BRAKE PIN	1	
8	MH014936900	HEX. HEAD NUT	2	3/8"-16UNF
9	MH15026400	PLAIN WASHER	4	
10	MH870001359	LOCK NUT	1	
11	MH992275379	HEX HEAD BOLT	1	M10 X 50LG.
12	MH992761014	HEX. NUT	1	M10
13	MH992275297	HEX HEAD BOLT	1	M8 X 30LG.
14	MH992781624	PLAIN WASHER	2	
15	MH992761013	HEX. NUT	1	M8
16	MH992826232	SPLIT PIN.	1	Ø3 X 32LG
17	MH60930300	HEX HEAD BOLT	2	3/8"-16UNF- 3-1/4 LG
18	MH798205064	BUSH	1	

- MH014607000 Rod End Pin 1

Brake Spares related to Drive Axle

1	MH4049618	Brake left	1	
2	MH40449624	Brake right	1	
3	MH40497021	(Brake Shoe) Liner	4	
4	MH40497041	Wheel Cylinder	2	

BRAKE TUBE ASSEMBLY



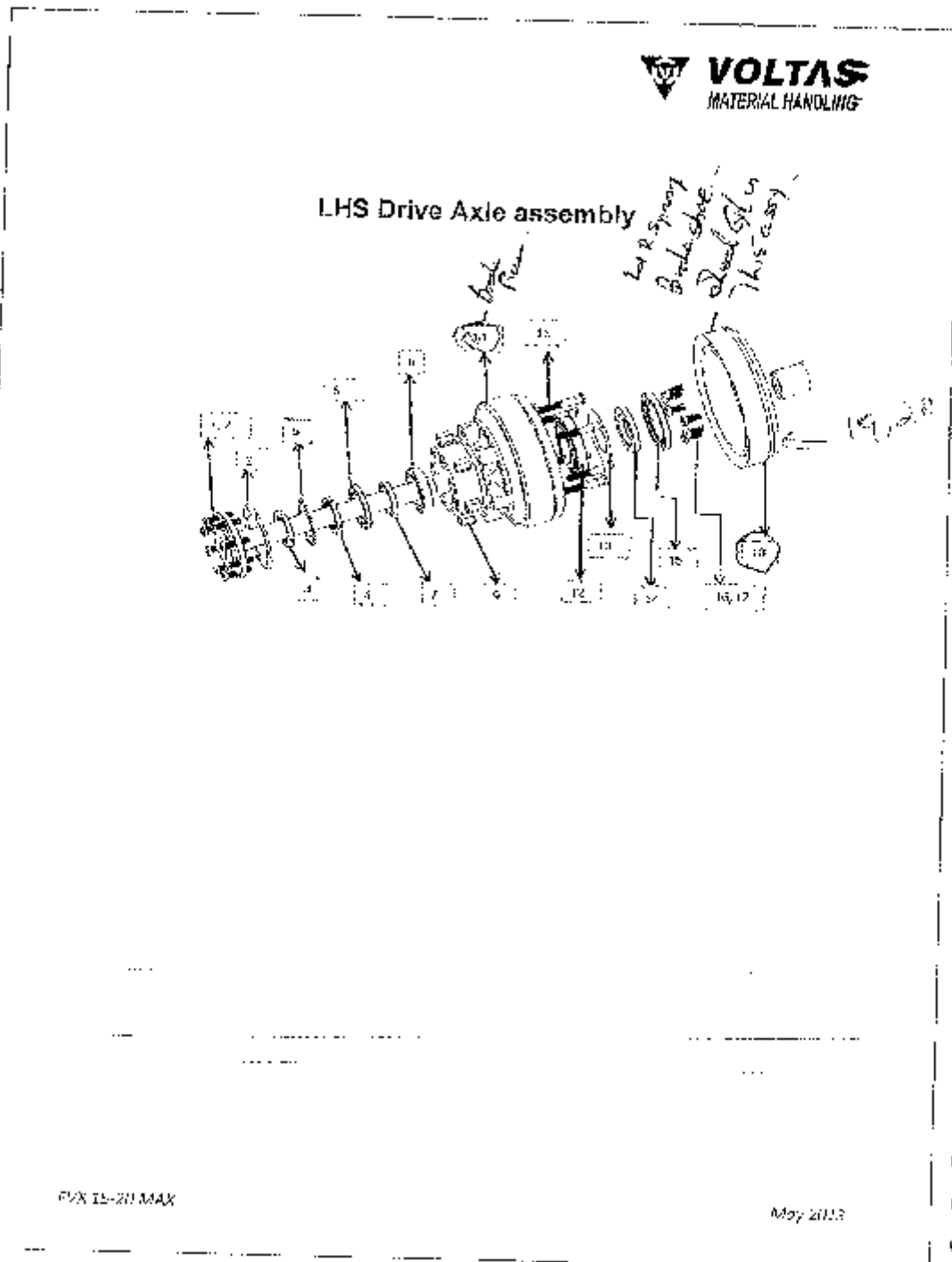
BRAKE TUBE ASSEMBLY

SR. NO.	PART NO.	DESCRIPTION	QTY	REMARKS
1	MH870001211	MASTER CYLINDER	1	
2	MH870001394	BANJO ADAPTOR	1	
3	MH08782	BRAKE HOSE	1	
4	*MH013370100	TEE	1	
5	MH799214043	LH BRAKE TUBE	1	
6	MH799214059	RH BRAKE TUBE	1	

*MH09947

Brake TEE Fitting

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LHS Drive Axle assembly

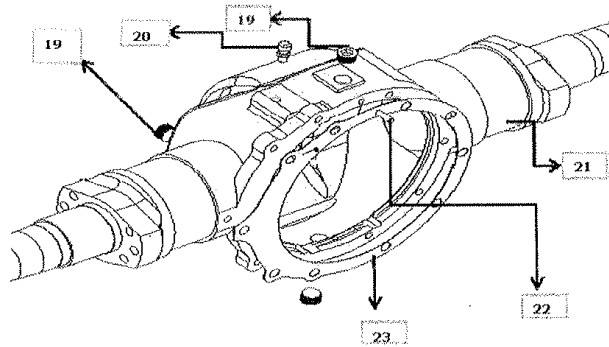
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
1	MH4049601	Hex head screw	16	
2	Mh4049602	Spring washer	16	
3	MH4049603	Gasket	2	
4	MH4049604	Nut	4	
5	MH4049605	Lock washer for nut	2	
6	MH4049606	Oil seal	2	
7	MH4049607	Bush, oil seal	2	
8	MH4049608	Taper roller bearing	2	
9	MH4049609	Wheel nut	12	
10	MH4049610	Wheel hub	2	
11	MH4049611	Wheel stud	12	
12	MH4049612	Taper roller bearing	2	
13	MH4049613	Anti-oil plate	2	
14	MH4049614	Bush, oil seal	2	
15	MH4049615	Oil seal	2	
16	MH4049616	Hex head screw	12	
17	MH4049617	Spring washer	12	
18	MH4049618	Brake, left	1	

19 MH42145021 Brake shoes 4

20 MH42145041 Wheel Cylinder 2



Axle Housing Center peice



EVX 15-20 MAX

May 2013



Axle Housing Center peice

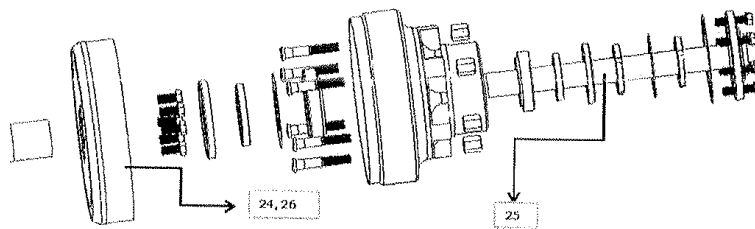
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
19	MH4049619	Plug	3	
20	MH4049620	Air breather plug	1	
21	MH4049621	Case, axle	1	
22	MH4049622	Taper pin	1	
23	MH4049623	Gasket	1	

EVX 15-20 MAX

May 2013



RHS Drive Axle Assembly



EVX 15-20 MAX

May 2013



RHS Drive Axle Assembly

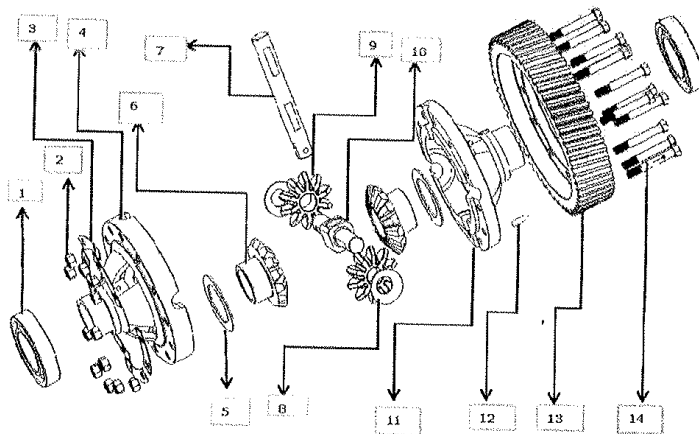
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
24	MH40449624	Brake, right	1	
25	MH40449625	Half shaft	2	
26	MH40449626	Cable, brake	2	

EVX 15-20 MAX

May 2013



Differential Assembly



EVX 15-20 MAX

May 2013

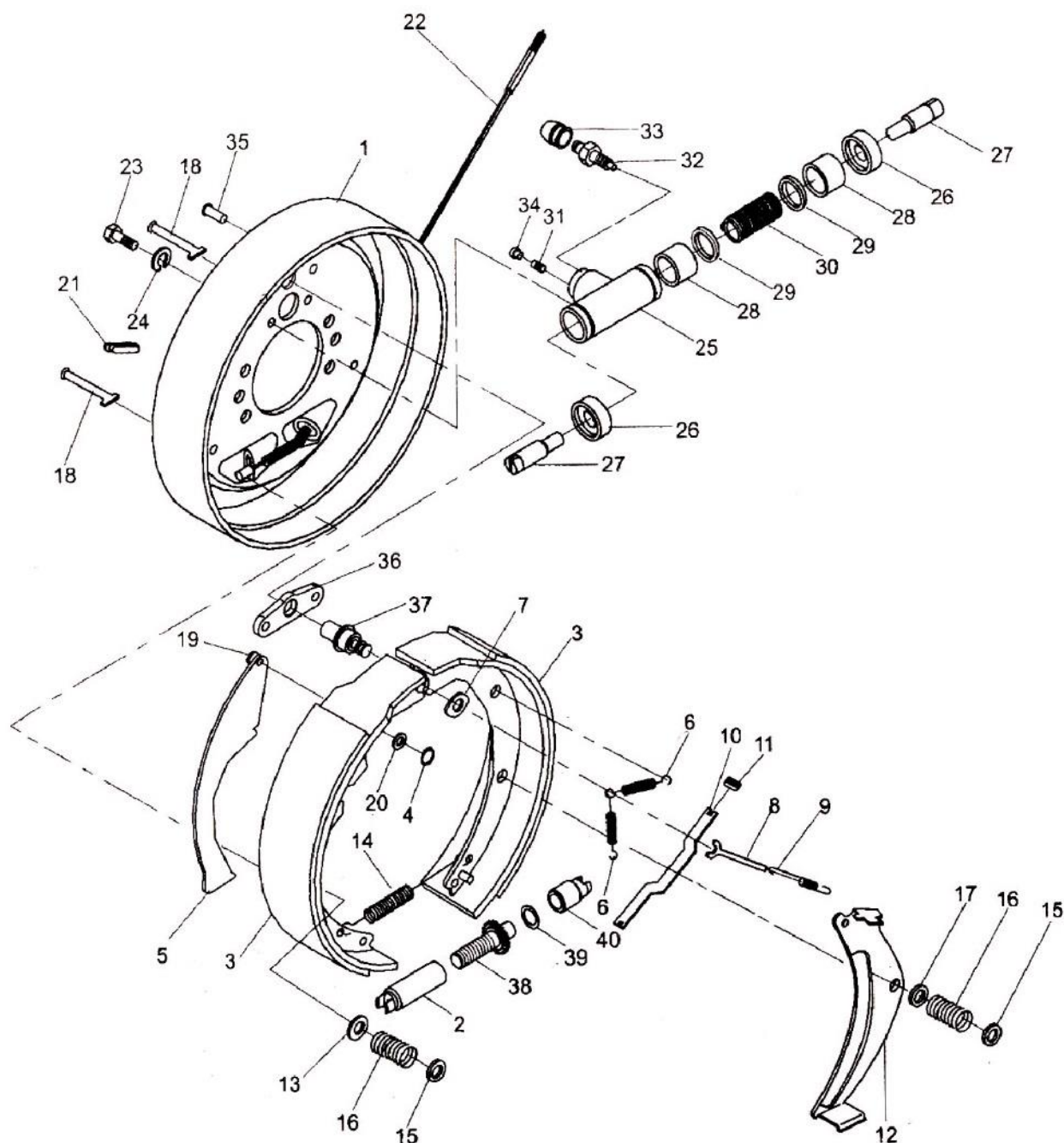


Differential Assembly

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
1	MH4049501	Deep Groove ball bearing	2	
2	MH4049502	Nut	12	
3	MH4049503	Anti-loose plate	6	
4	MH4049504	Case, cross joint (L.H.)	1	
5	MH4049505	Washer	2	Selective
	MH4049506	Washer	2	Selective
	MH4049507	Washer	2	Selective
	MH4049508	Washer	2	Selective
6	MH4049509	Gear, side	2	
7	MH4049510	Shaft, pinion II	1	
8	MH4049511	Washer	4	Selective
	MH4049512	Washer	4	Selective
	MH4049513	Washer	4	Selective
9	MH4049514	Gear, pinion	4	
10	MH4049515	Shaft, pinion I	1	
11	MH4049516	Case, cross joint (R.H.)	1	
12	MH4049517	Taper pin	1	
13	MH4049518	Ring Gear	1	
14	MH4049519	Bolt for ring gear	12	

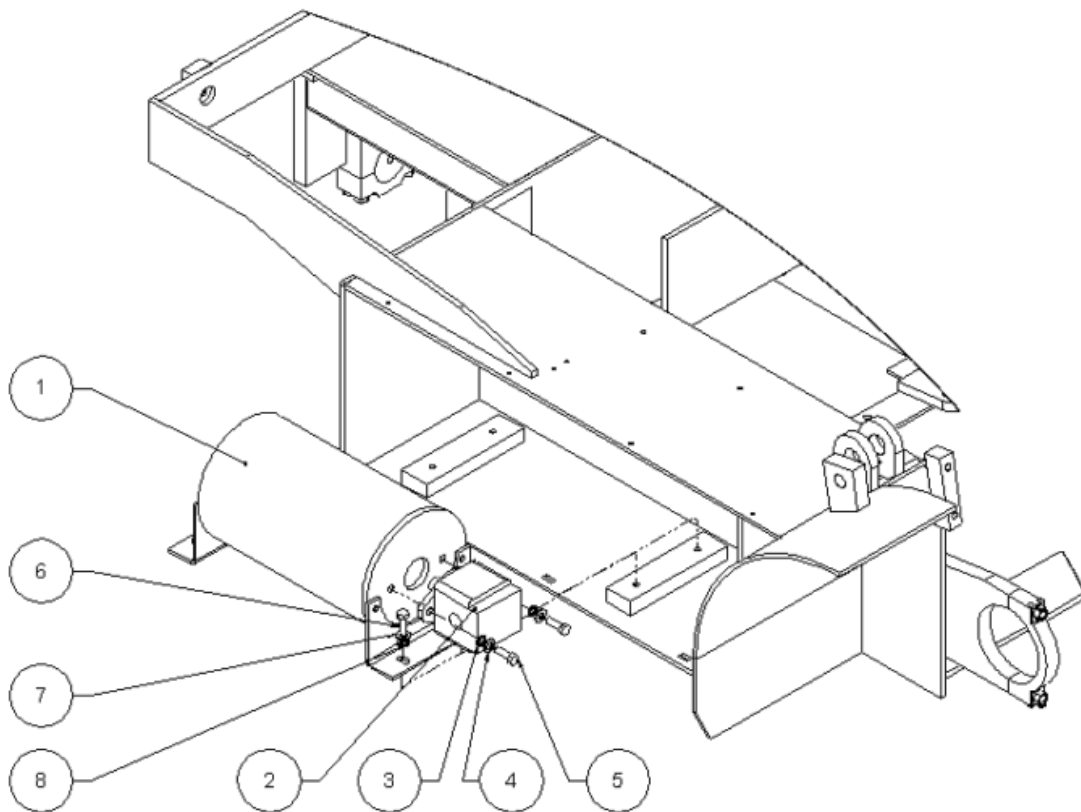
EVX 15-20 MAX

May 2013



1	MH4049618	Brake left	1	
2	MH40449624	Brake right	1	
3	MH40497021	(Brake Shoe) Liner	4	
4	MH40497041	Wheel Cylinder	2	

PUMP & MOTOR ASSEMBLY⁸⁷⁰⁰⁰¹⁷⁵⁴



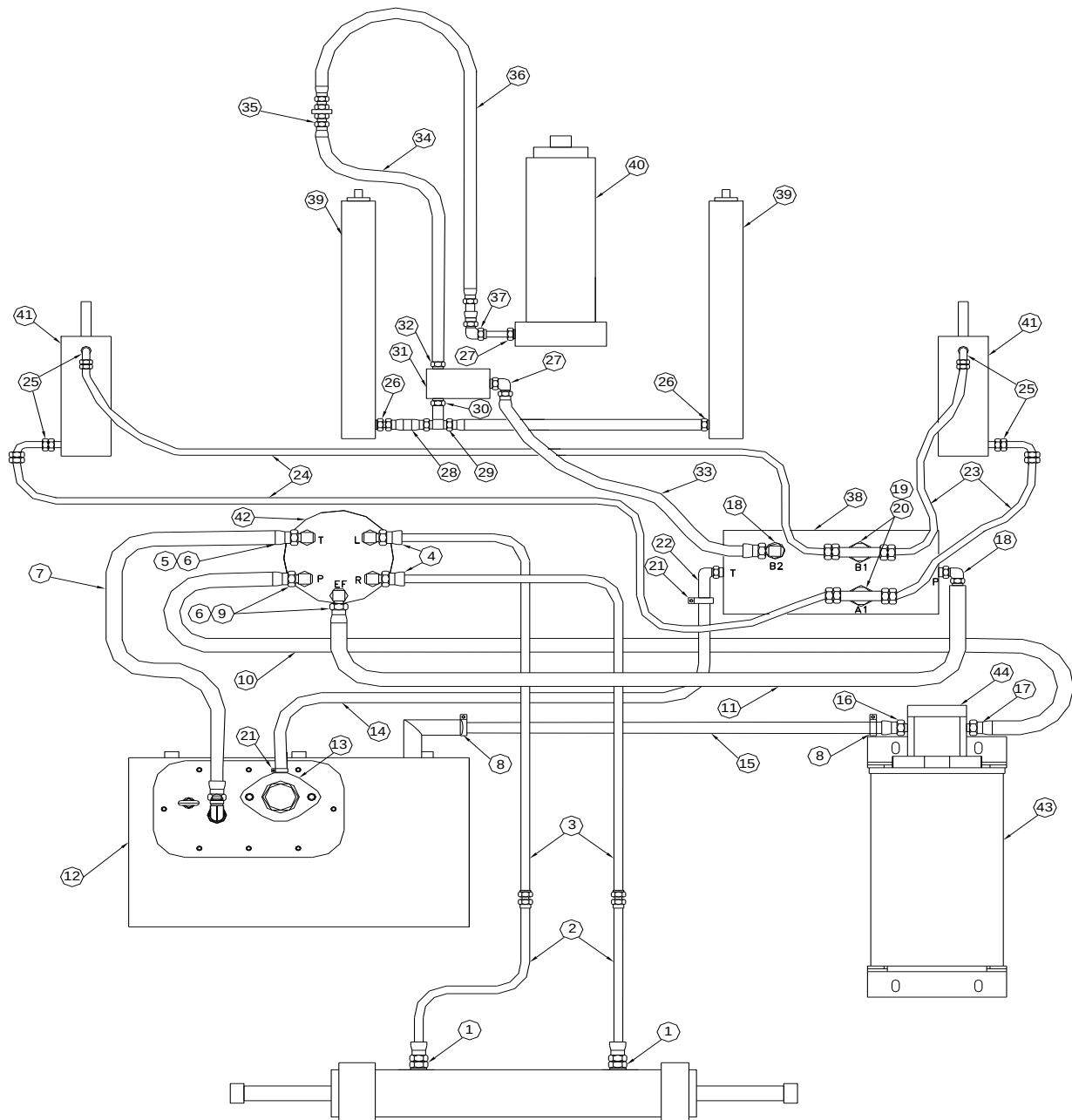
DC HYD MOTOR & PUMP ASSEMBLY

ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH 40519	DC PUMP MOTOR 8KW/48V	1	
2	MH 17747	HYDRAULIC PUMP Dowty	1	
3	MH 992785634	PLAIN WASHER	2	M10
4	MH 992791535	SPRING WASHER	2	M10
5	MH 992571377	HEX. SOCKET HEAD CAP SCREW	2	M10 X 30LG.
6	MH 992785634	PLAIN WASHER	2	M10
7	MH 992791535	SPRING WASHER	2	M10
8	MH 992275376	HEX HEAD BOLT	1	M10 X 25LG.

9	MH17552A009	Carbon Brush Hoist DC Motor	4	
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AC HYD MOTOR & PUMP ASSEMBLY

ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH42180	AC PUMP MOTOR 10KW 48V (PM)	1	
2	MH17571	HYDRAULIC PUMP DOWTY	1	
3	MH992785634	PLAIN WASHER	2	M10
4	MH992791535	SPRING WASHER	2	M10
5	MH992571377	HEX. SOCKET HEAD CAP SCREW	2	M10 X 30LG.
6	MH992785634	PLAIN WASHER	2	M10
7	MH992791535	SPRING WASHER	2	M10
8	MH992275376	HEX HEAD BOLT	1	M10 X 25LG.



HYDRAULIC CIRCUIT

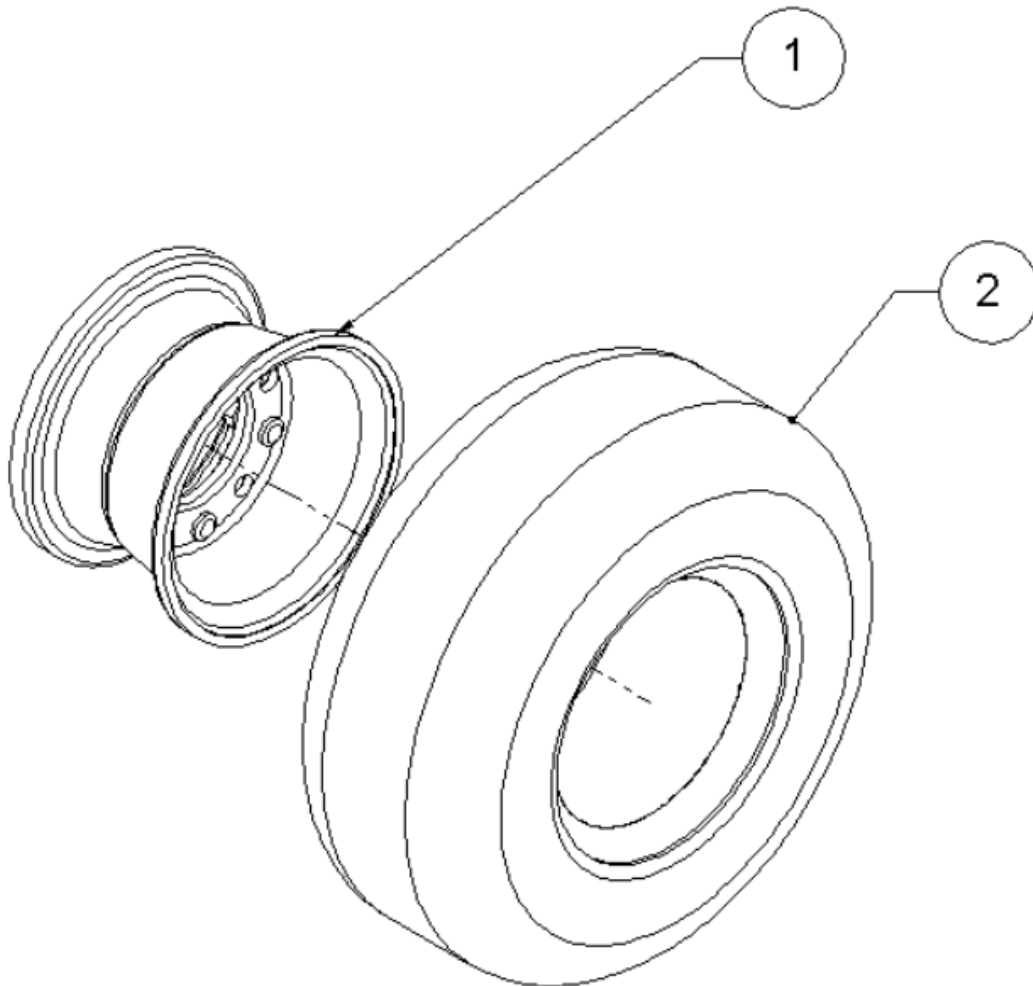
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
1	MH993750109	STRAIGHT ADAPTOR	2	
2	MH799209063	HYDRUALIC BRAKE TUBE	2	
3	MH993811160	HOSE (1600 LG.)	2	1/4" R1
4	MH11193	STRAIGHT ADAPTOR	2	
5	MH11332	STRAIGHT ADAPTOR	1	
6	MH993750612	90 DEG. SWIVEL ELBOW	3	
7	MH799209184	HOSE (600 LG.)	1	3/8" R3
8	MH993728140	HOSE CLAMP	2	
9	MH11304	STRAIGHT ADAPTOR	2	
10	MH799209174	HOSE (1100 LG.)	1	1/2" R2
11	MH799209194	HOSE (500 LG.)	1	1/2" R2
12	MH799209033	HYRAULIC TANK	1	
13	MH799806123	RETURN LINE FILTER	1	
14	MH003338400	BARE HOSE	1	3/4" R3
15	MH799806114	BARE HOSE	1	1" R3
16	MH797006444	STRAIGHT ADAPTOR	1	
17	MH42098	STRAIGHT ADAPTOR	1	
18	MH00896	STRAIGHT ADAPTOR	2	
19	MH08311	STRAIGHT ADAPTOR	2	
20	MH44Z487D5	TEE FITTING	2	
21	MH993728130	HOSE CLAMP	2	
22	MH42111	90 DEG. ELBOW	1	
23	MH993811055	HOSE (550 LG.)	2	1/4" R1
24	MH993811075	HOSE (750 LG.)	2	1/4" R1
25	MH870001109	90 DEG. SWIVEL ELBOW	4	
26	MH00888	90 DEG. SWIVEL ELBOW	2	
27	MH40680	90 DEG. ELBOW	1	
28	MH17563	HOSE ASSY 3/4 UNF 90 D BEND	1	
29	MH12698	HOSE	1	
30	MH12619	TEE FITTING	1	
31	MH12319	MANIFOLD	1	
32	MH12648	STRAIGHT ADAPTOR	1	

33	MH799201204	HOSE (550 LG.)	1	1/2" R2
34	MH14557	HOSE	1	
35	12519	STRAIGHT FITTING	1	

HYDRAULIC CIRCUIT

ITEM NO.	PART NUMBER	DESCRIPTION	QTY	REMARKS
36	MH18319	HOSE	1	
37	MH42086	90 DEG. SWIVEL ELBOW	1	
38	MH42052	2 SPOOL CONTROL VALVE	1	
39	**MH16208	SIDE CYLINDER	2	
40	***MH16201	CENTER CYLINDER	1	
41	*MH40504	TILT CYLINDER	2	
42	MH42046	STEERING UNIT	1	
43	MH42180	PUMP MOTOR 10KW 48V (PM)	1	
44	MH17571	HYDRAULIC PUMP	1	
	*MH42443	Tilt Cyl (HVT) Sundaram	1	
	*MH42031	Tilt Cyl(HVT) DC12398	2	
	**MH42482	Side Lift Cyl RAM type	2	
	***MH42238	Centre Cyl Dantal	1	
			1	

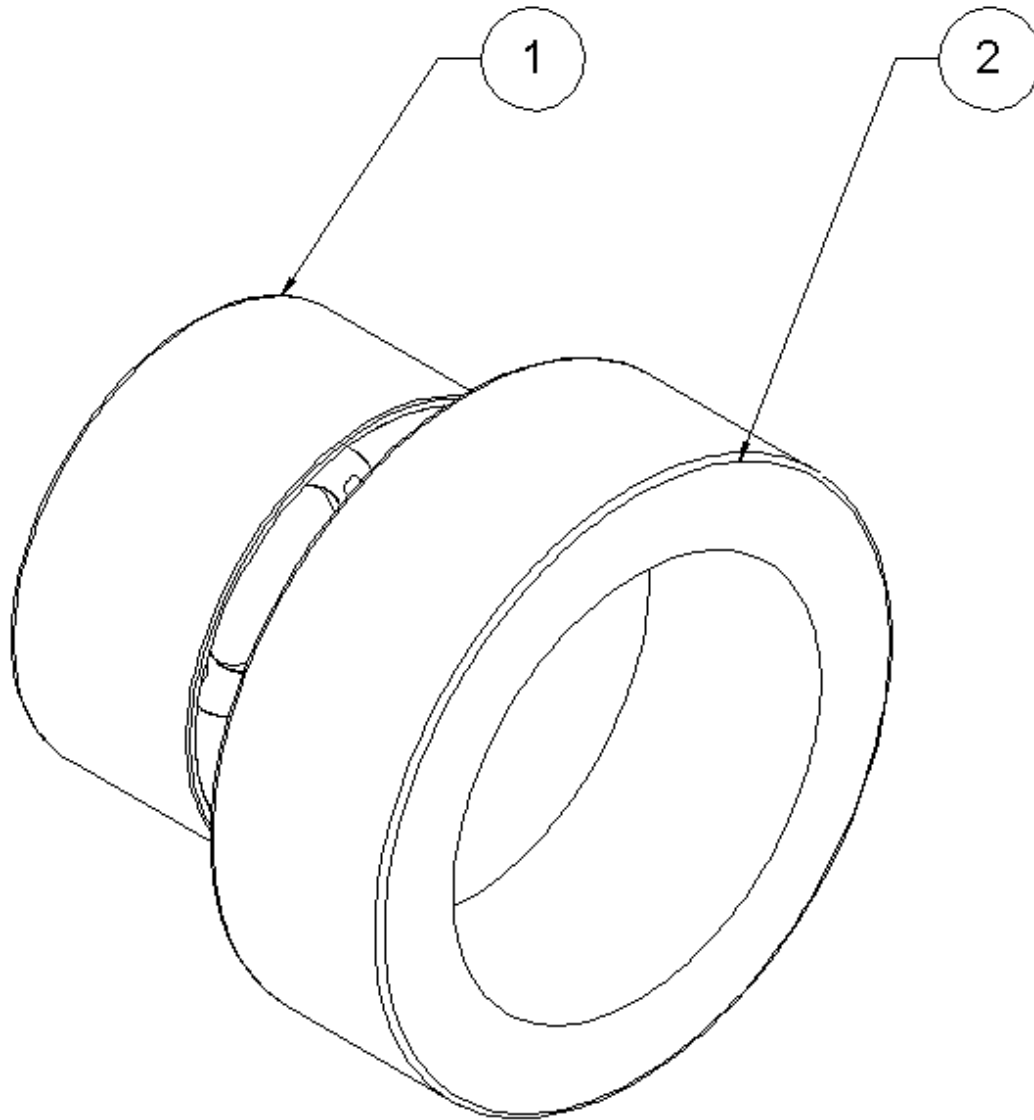
REAR TYRE ASSEMBLY



REAR TYRE & RIM ASSY(MH79820422)

ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH798204032	Spilt RIM 4.33''*8''	2	
2	MH798204054	TYRE ASSEMBLY 18x7-8	2	

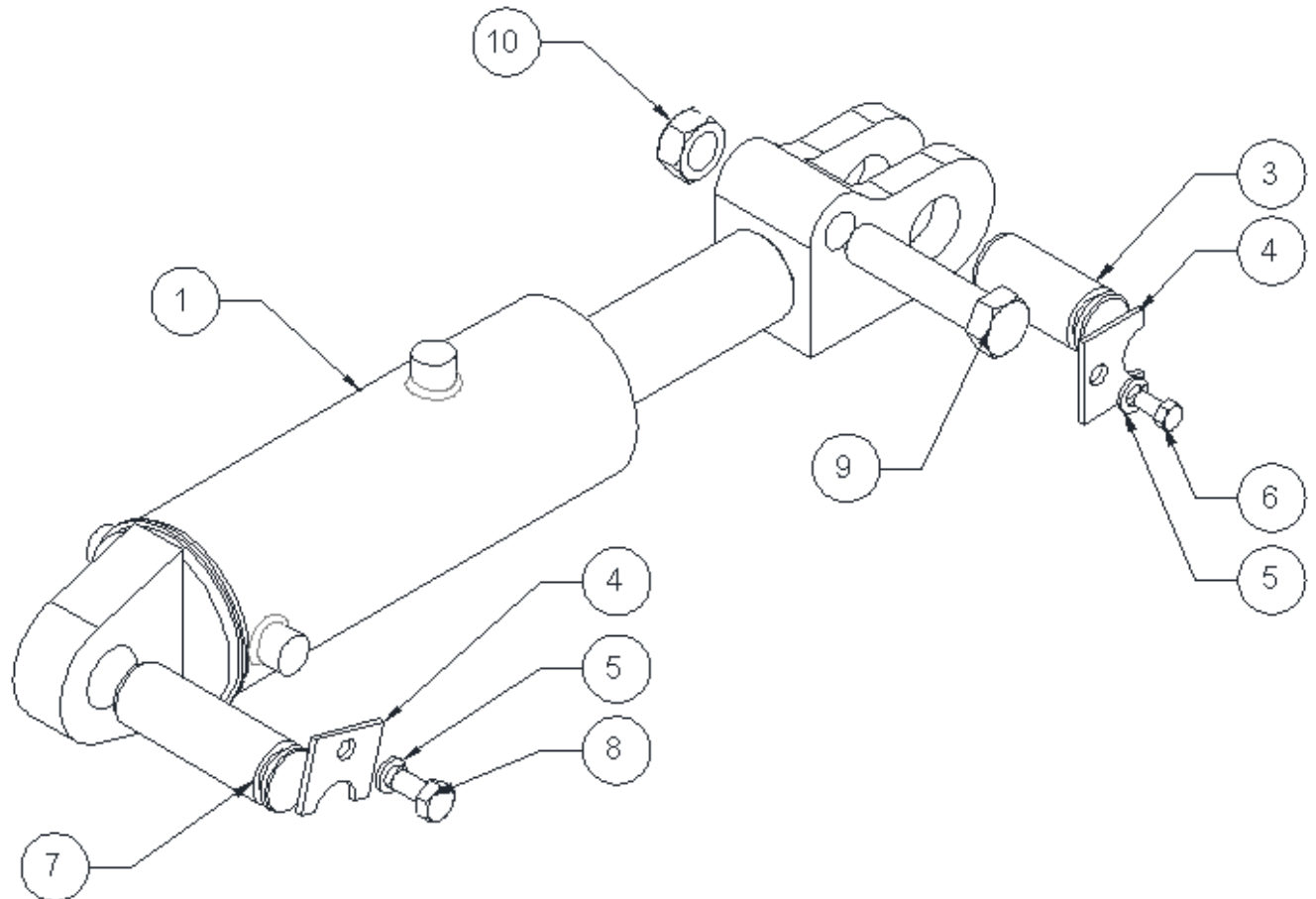
FRONT TYRE ASSEMBLY



FRONT TYRE & RIM ASSY(MH799212063)

ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH799212033	FRONT RIM for 21" press on Type	2	
2	MH799212044	TYRE PRESS ON 21x7x15 Press on Type	2	

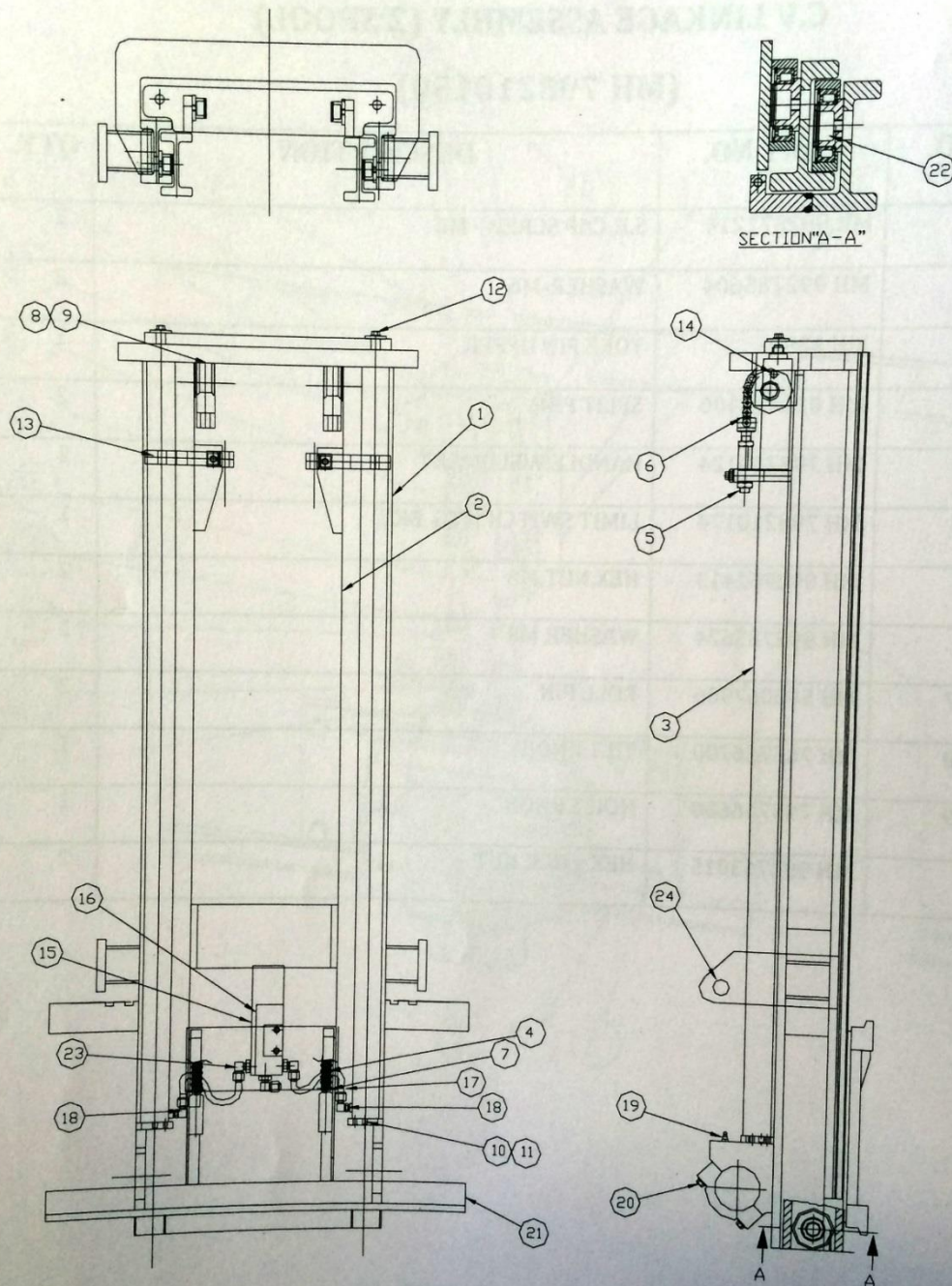
TILT CYLINDER ASSEMBLY



ITEM NO	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH40503	TILT CYLINDER	1	HVM
	MH42031	TILT CYLINDER	1	HVT
2	MH501900300	ROD END	1	
3	MH509816101	Pin TILT CYL. (common)	1	
4	MH502658100	PIN RETAINER	2	
5	MH992791535	SPRING WASHER FOR M10	2	
6	MH55486700	HEX. HEAD BOLT	1	3/8" X 5/16" LG. GR-5
7	MH796414049	Rear PIN-TILT CYLINDER	1	
8	MH992275374	HEX. HEAD BOLT	1	M10x16 LG.
9	MH63894200	HEX. HEAD BOLT	1	3/4" X 3 1/4" LG. GR-5
10	MH14939500	HEX NUT	1	3/4" x 10 UNC

TILT CYLINDER ASSEMBLY

VOLTAS



**HIGH VISIBILITY MAST ASSEMBLY
(MH 798211021)**

HVM MAST ASSEMBLY

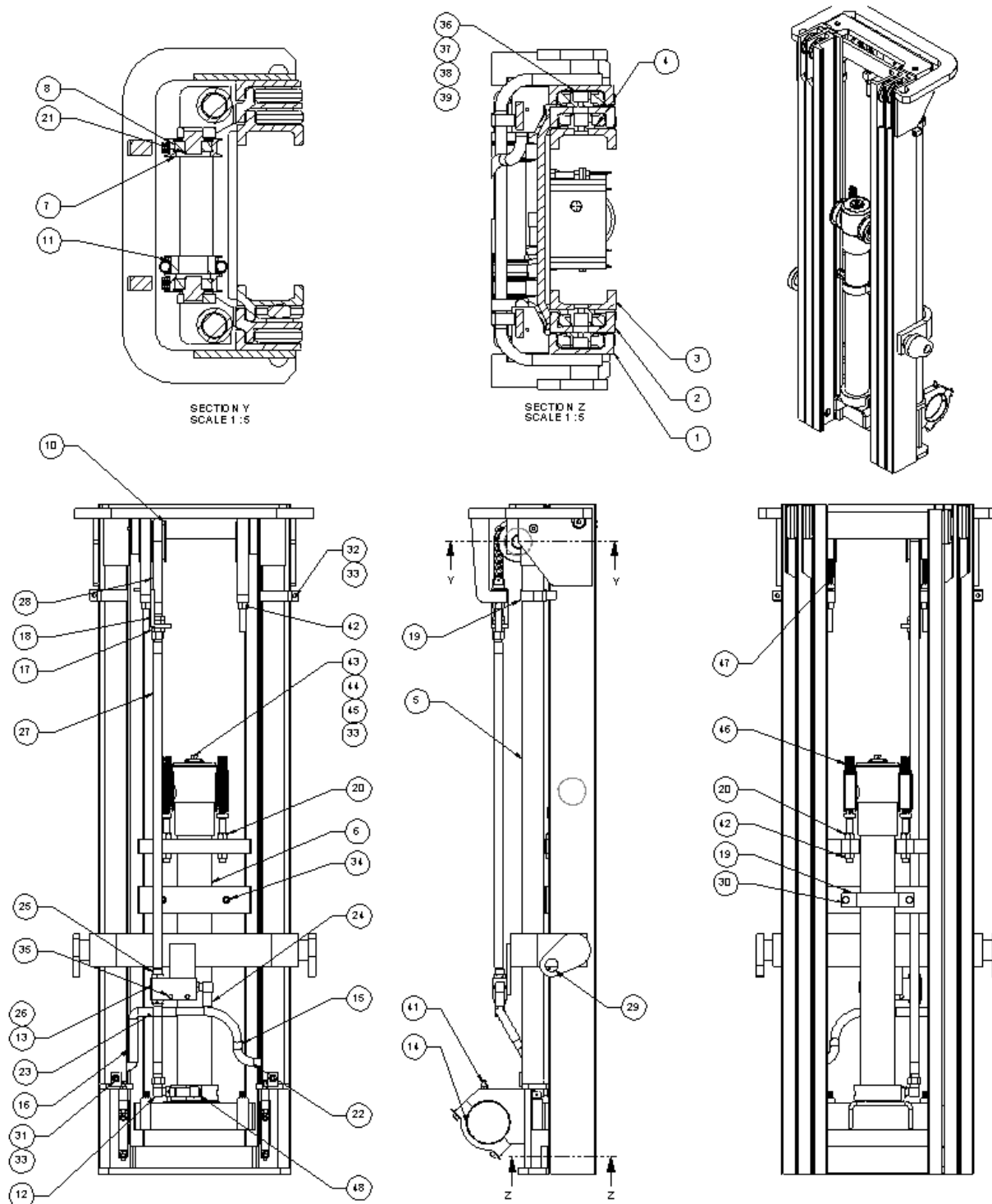
ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH799206532	STATIONARY WELDMENT	1	
2	MH799206492	ELEVATING WELDMENT	1	
3	*MH16200	SIDE LIFT CYL RAM DC 6040 Sim MH16018(Wipro)	2	
	MH42498	LIFT CYL (Sundaram make)		
4	-	-		
5	MH785769001	NYLOCK NUT	2	
6	MH10135	CARRIAGE LIFTING CHAIN . (118 LINKS)	2	
8	-	-		
9	MH500353903	BEARING	2	
10	MH992821555	HARDENED CYL PIN	2	
11	-	-		
12	MH992175382	HEX HEAD SCREW M10*45LG	2	
13	MH799226854	CYL CLAMP	2	
14	MH508623301	CHAIN SHEEVE COMMON	2	
15	MH12319	MANIFOLD	1	
16	MH13614	FLOW REGULATOR(2 HOLE)	1	
17	-	-		
18	-	-		

19	MH799229438	HOSES & FIITINGS	1	
	MH785324300	NIPPLE	2	
20	MH512505100	LOCK PLATE	4	
21	MH799227072	CARRIGE ASSY	1	
22	MH504562101	WEAR PLUG W/Disc	4	
	MH42084	INTEGRAL BEARING EQV MH870001754	4	
23		ELBOW	1	
24	MH799226284	BUSHING	4	

Mh799217123

Phos bronze bush for mast

HVT 2125 MAST ASSEMBLY

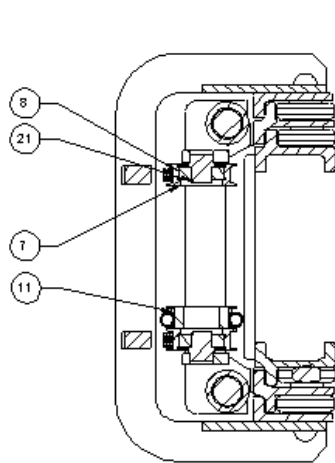


HVT 2125 MAST ASSEMBLY

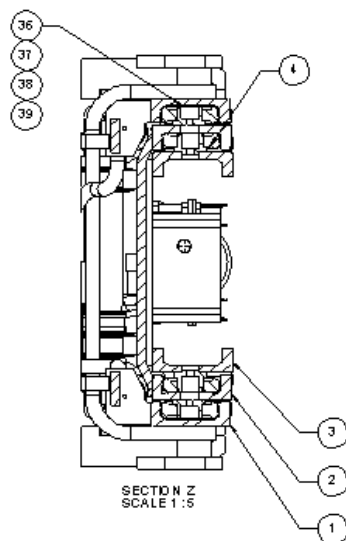
ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH799206422	STATIONARY WELDMENT	1	
2	MH799206252	SECONDARY WELDMENT	1	
3	MH799206232	PRIMARY ELEVATING WELDMENT	1	
4	*MH870001754	INTEGRAL BEARING	8	
5	MH42482	SIDE HOIST CYL. ASSLY	2	
6	MH42238	Central Cyl Dantal DC12756.	1	
7	MH17295	SHEAVE Chain	4	
8	MH500353903	BEARING	4	
9	MH797930614	SHEAVE SUPPORT WELDMENT	1	
10	MH17321	HOSE SHEAVE	1	
11	MH12521	BUSHING	1	
12	MH42086	90 DEG ELBOW	1	
13	MH12139	MANIFOLD	1	
14	**MH799206103	MAST BUSHING	2	
15	MH12698	HOSE	1	
16	MH17563	HOSE ASSY LH	1	
17	MH12519	STRAIGHT FITTING	1	
18	MH799206564	HOSE TAKE-UP BRACKET	1	
19	MH797930584	CYL. CLAMP	2	
20	MH014937500	NYLOCK NUT	4	
21	MH060379000	EXT. CIRCLIP	4	
22	MH00888	90 DEG. SWIVEL ELBOW	2	
23	MH12619	TEE FITTING	1	
24	MH721552700	90 DEG ELBOW	1	
25	MH12648	STRAIGHT FITTING	1	
26	MH10106	FLOW REGULATOR 4 HOLES	2	
27	MH14557	HOSE	1	
28	MH18319	HOSE	1	
29	MH508990100	TILT PIVOT BUSHING	2	
30	MH992275433	HEX. HD. SCREW	2	M12x100 LG.
31	MH992275377	HEX SCREW	2	M10x30 LG.
32	MH992275370	HEX. HD. BOLT	2	M10x30 LG.
33	MH992791535	SPRING WASHER	7	M10
34	MH992761115	HEX. HD. NUT	6	M10
35	MH992176386	HEX. HD. BOLT	2	M10x65 LG.

HVT 2125 MAST ASSEMBLY

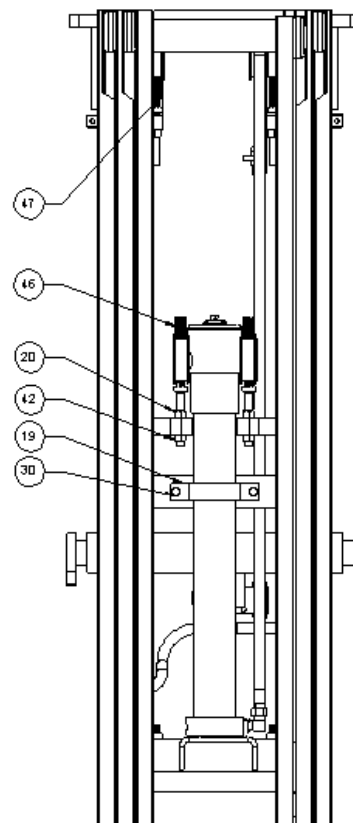
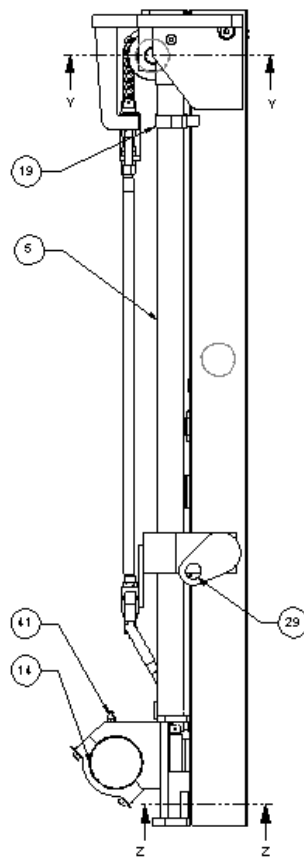
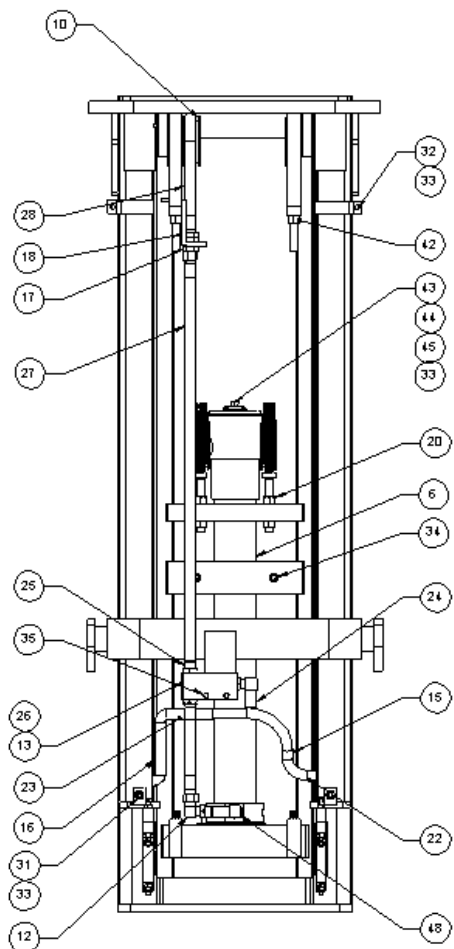
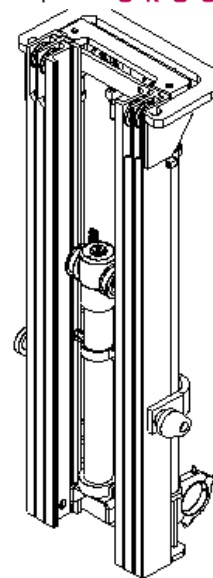
ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
36	MH504562101	WEAR PLUG	8	
37	MH504560100	DISC - WEAR PLUG	8	
38	MH992573419	GRUB SCREW	8	M12 X 35 LG.
39	MH992761015	HEX NUT	8	1/2"
40	MH785324300	LUBE FITTING	2	
41	MH992551288	SOC. HD. SCREW	1	5/8" UNFx7/8" LG.
42	MH785769001	NYLOCK NUT	4	
43	MH513837100	SPHERICAL WASHER	1	
44	MH992246170	HEX. HD. SCREW	1	3/8" UNCx1 1/4" LG.
45	MH11493	SPACER	3	
46	MH13344	CHAIN ASSY.	2	76 LINKS
47	MH515924419	Pre Lifting Chain		103link
48	MH40680	90 DEG. ELBOW	1	
	*MH42084	Integral bearing eqv MH870001754	1	
	**MH799217123	Phos Bronze Bush for Mast	2	
	MH504560100	Sheave Chain (Common)	3	
	MH799209608	Hoses & Fitting Kit HVT2125	1	
			8	
			4	
			1	
			2	



SECTION Y
SCALE 1:5



SECTION X Z
SCALE 1:5



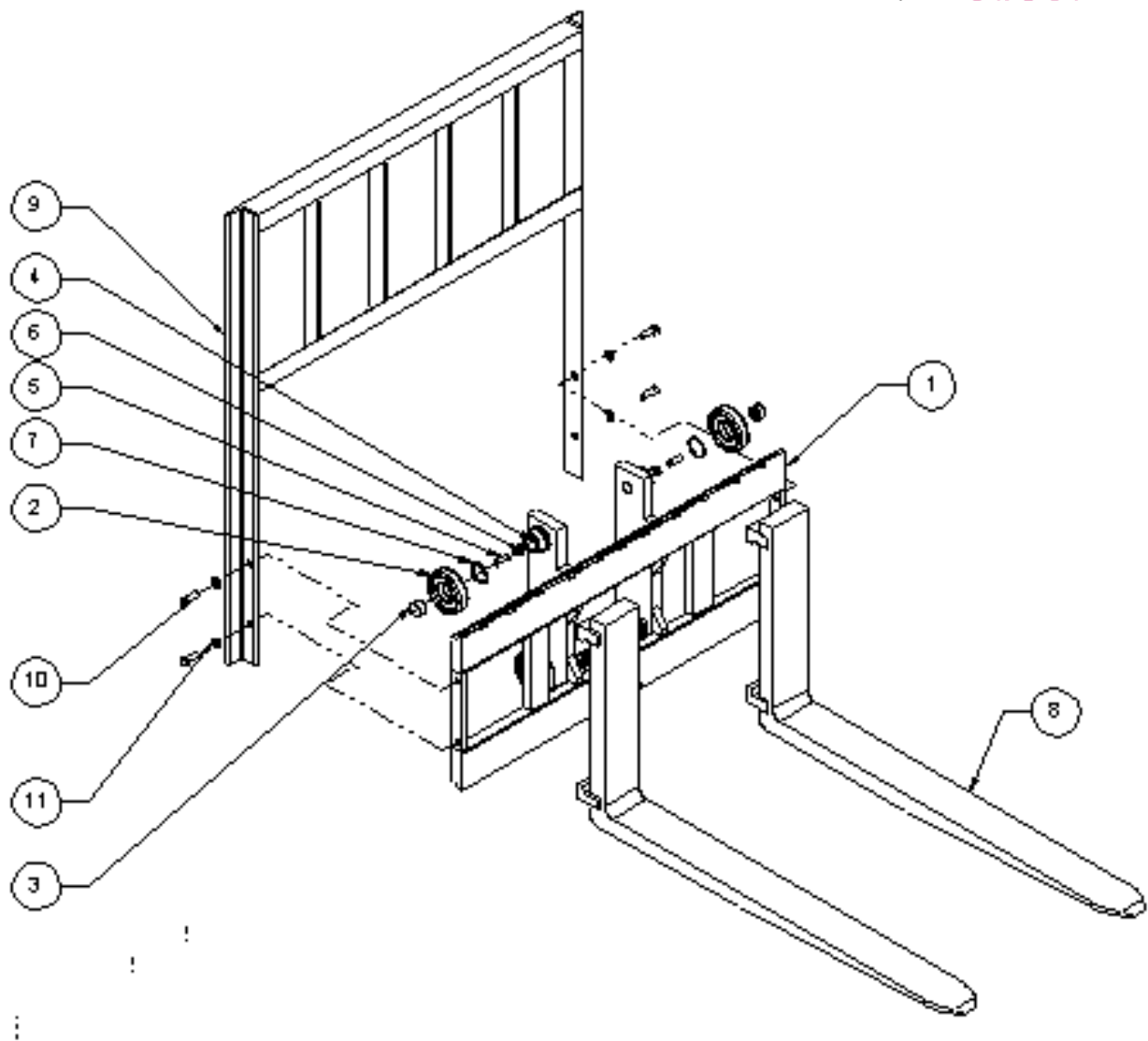
HVT 2700 MAST ASSEMBLY

ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
1	MH799206612	STATIONARY WELDMENT	1	
2	MH799206652	SECONDARY WELDMENT	1	
3	MH799206662	PRIMARY ELEVATING WELDMENT	1	
4	*MH870001754	INTEGRAL BEARING	8	
5	MH14635/ MH16206	SIDE HOIST CYL. ASSLY	2	
6	MH15995	Central Cyl	1	
7	MH17295	SHEAVE Chain	4	
8	MH500353903	BEARING	4	
9	MH797930614	SHEAVE SUPPORT WELDMENT	1	
10	MH17321	HOSE SHEAVE	1	
11	MH12521	BUSHING	1	
12	MH42086	90 DEG ELBOW	1	
13	MH12139	MANIFOLD	1	
14	**MH799206103	MAST BUSHING	2	
15	MH12698	HOSE	1	
16	MH17563	HOSE ASSY LH	1	
17	MH12519	STRAIGHT FITTING	1	
18	MH799206564	HOSE TAKE-UP BRACKET	1	
19	MH797930584	CYL. CLAMP	2	
20	MH014937500	NYLOCK NUT	4	
21	MH060379000	EXT. CIRCLIP	4	
22	MH00888	90 DEG. SWIVEL ELBOW	2	
23	MH12619	TEE FITTING	1	
24	MH721552700	90 DEG ELBOW	1	
25	MH12648	STRAIGHT FITTING	1	
26	MH10106	FLOW REGULATOR 4 HOLES	2	
27	MH14557	HOSE	1	

28	MH18319	HOSE	1	
29	MH508990100	TILT PIVOT BUSHING	2	
30	MH992275433	HEX. HD. SCREW	2	M12x100 LG.
31	MH992275377	HEX SCREW	2	M10x30 LG.
32	MH992275370	HEX. HD. BOLT	2	M10x30 LG.
33	MH992791535	SPRING WASHER	7	M10
34	MH992761115	HEX. HD. NUT	6	M10
35	MH992176386	HEX. HD. BOLT	2	M10x65 LG.

HVT 2700 MAST ASSEMBLY

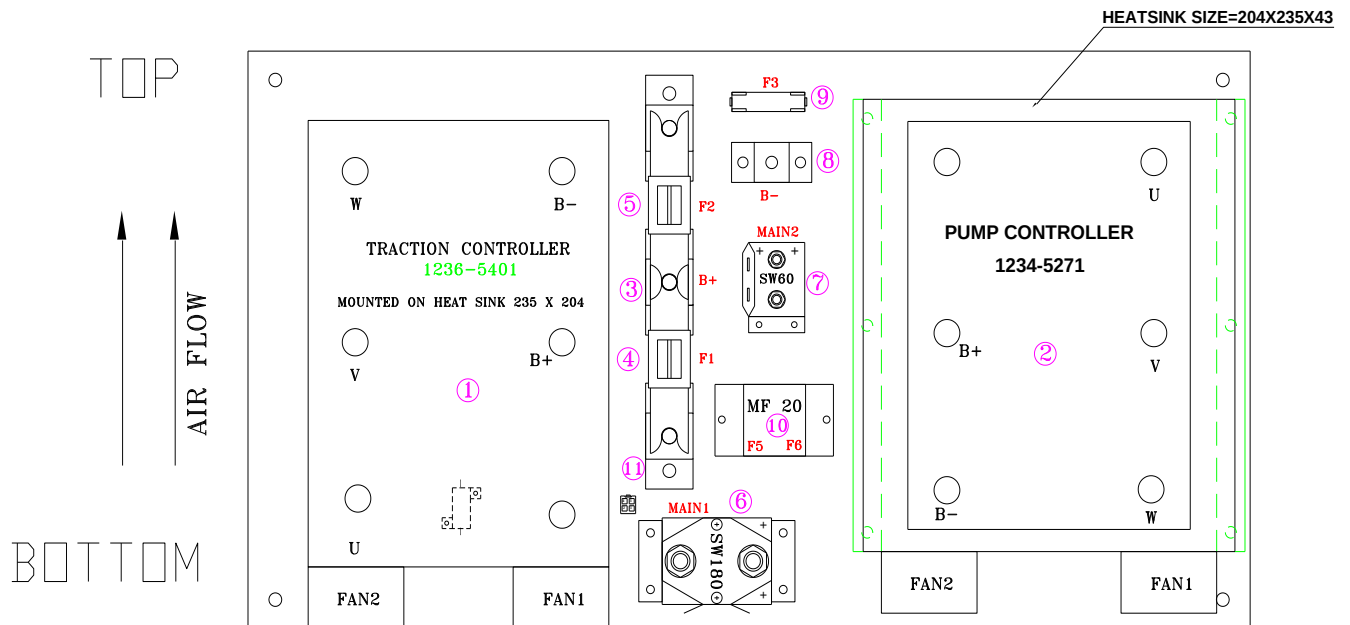
ITEM NO.	PART NO.	DESCRIPTION	QTY.	REMARKS
36	MH504562101	WEAR PLUG	8	
37	MH504560100	DISC - WEAR PLUG	8	
38	MH992573419	GRUB SCREW	8	M12 X 35 LG.
39	MH992761015	HEX NUT	8	1/2"
40	MH785324300	LUBE FITTING	2	
41	MH992551288	SOC. HD. SCREW	1	5/8" UNFx7/8" LG.
42	MH785769001	NYLOCK NUT	4	
43	MH513837100	SPHERICAL WASHER	1	
44	MH992246170	HEX. HD. SCREW	1	3/8" UNCx1 1/4" LG.
45	MH11493	SPACER	3	
46	MH13344	CHAIN ASSY.	2	76 LINKS
47	MH515924419	Pre Lifting Chain		103link
48	MH40680	90 DEG. ELBOW	1	
	*MH42084	Integral bearing eqv MH870001754	1	
	**MH799217123	Phos Bronze Bush for Mast	2	
	MH504560100	Sheave Chain (Common)	3	
	MH799209448	Hoses Kit	1	
			8	
			4	
			1	
			2	



CARRIAGE ASSEMBLY

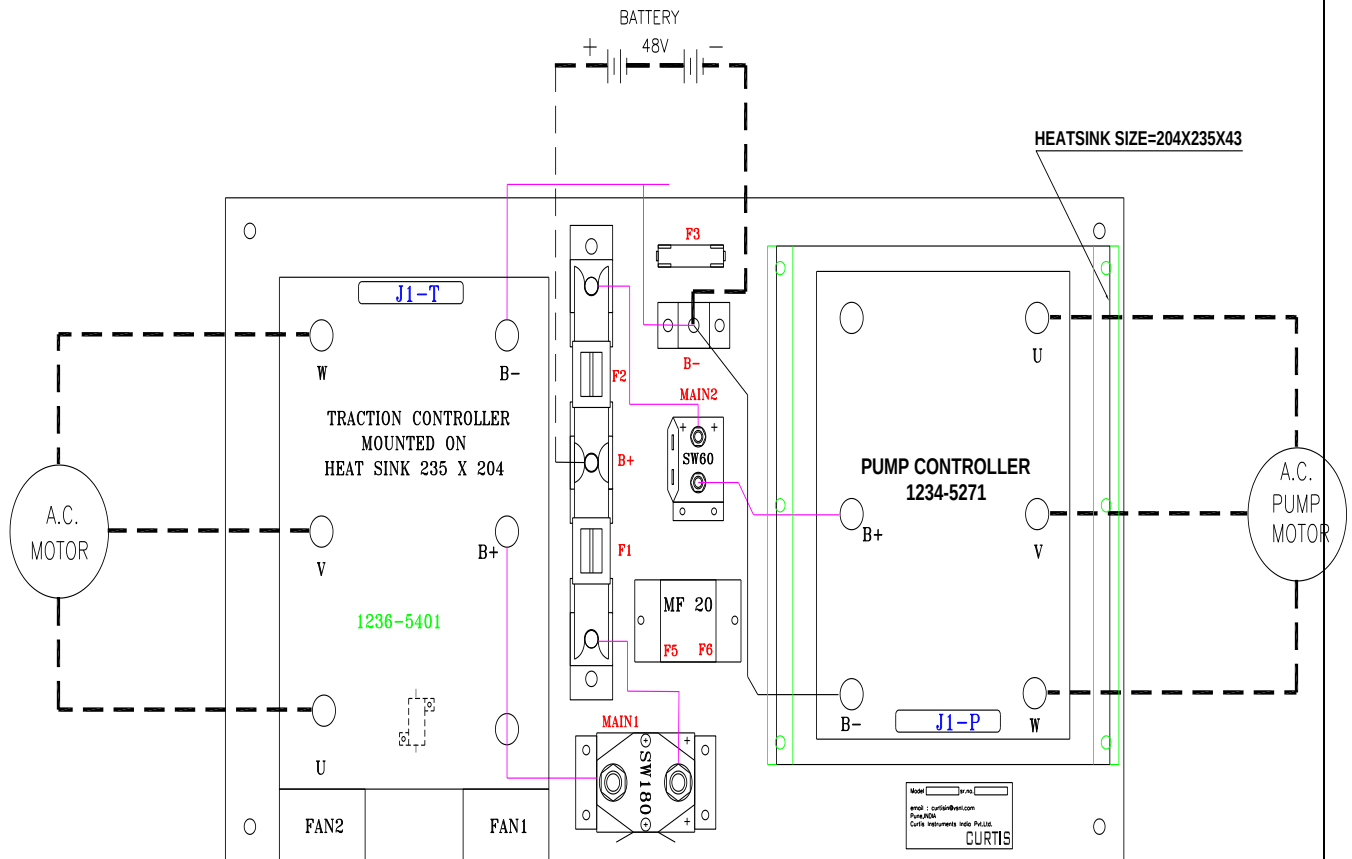
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	REMARKS
1	MH799207112	CARRIAGE WELD'T	1	
2	MH870001754	INTEGRAL BEARING	4	
3	MH504562101	WEAR PLUG	4	
4	MH504560100	DISC - WEAR PLUG	4	
5	MH992573419	GRUB SCREW	4	M12x35 LG.
6	MH992761015	HEX NUT	4	1/2"
7	MH60313900	CIRCLIP (EXTERNAL)	4	
8	MH11132	FORK	2	40x100x1070
9	MH799820042	LOAD BACK REST	1	
10	MH992175420	HEX.BOLT	4	M12x35 LG.
11	MH992791545	SPRING WASHER	4	M12

COMPONENT LAYOUT FOR EVX20MAX FULLY AC



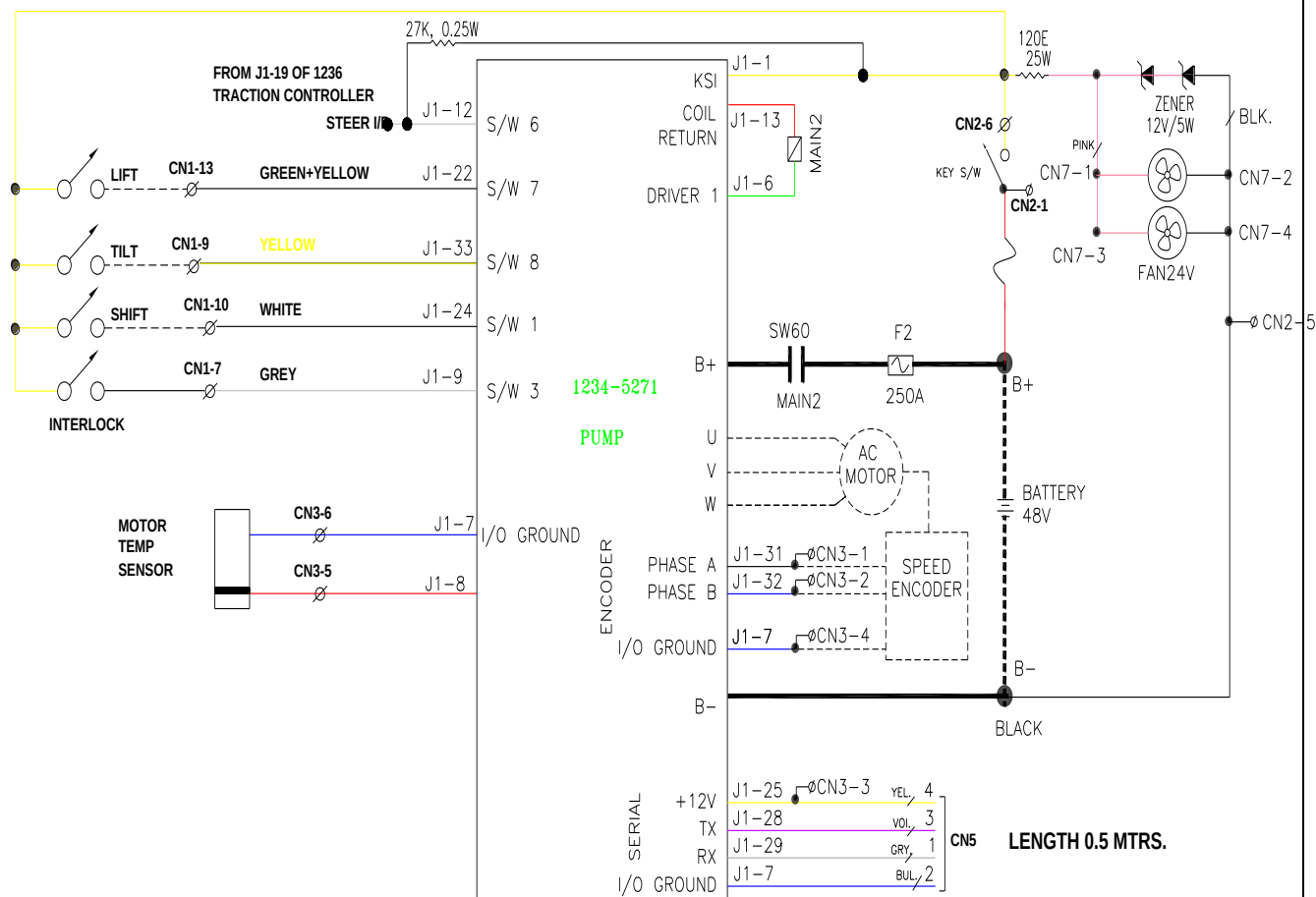
Component No.	Description
①	Controller 1236-5401
②	Controller 1234-5271
③	B+ Insulator
④	MAIN1 Fuse 250A
⑤	MAIN2 Fuse 250A
⑥	MAIN1 Contactor SW180
⑦	MAIN2 Contactor SW60
⑧	B- Insulator
⑨	Control Fuse 10A
⑩	Fuse Base MF-20
⑪	CN4 for fan connections

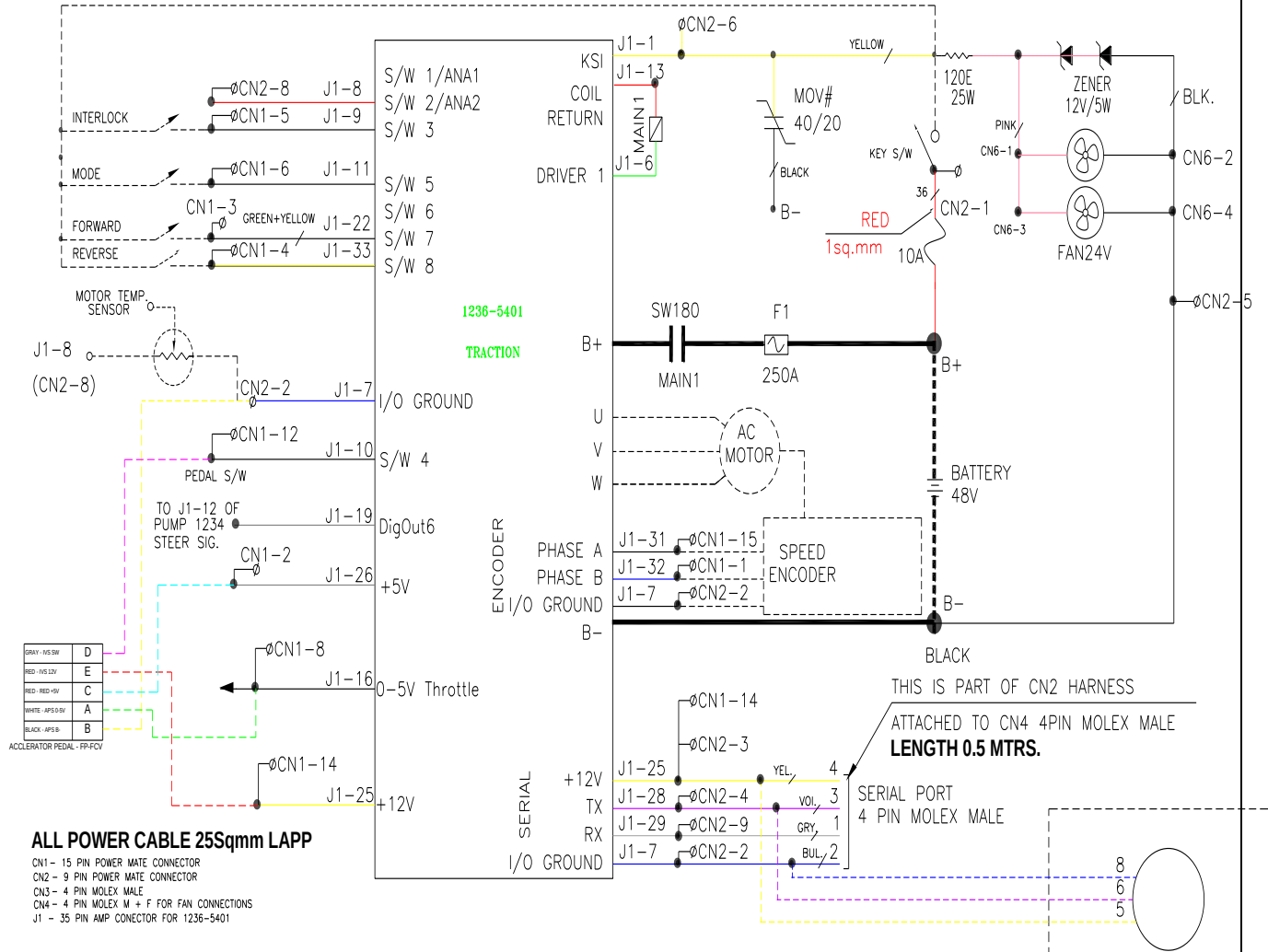
POWER WIRING DIAGRAM FOR EVX20MAX FULLY AC

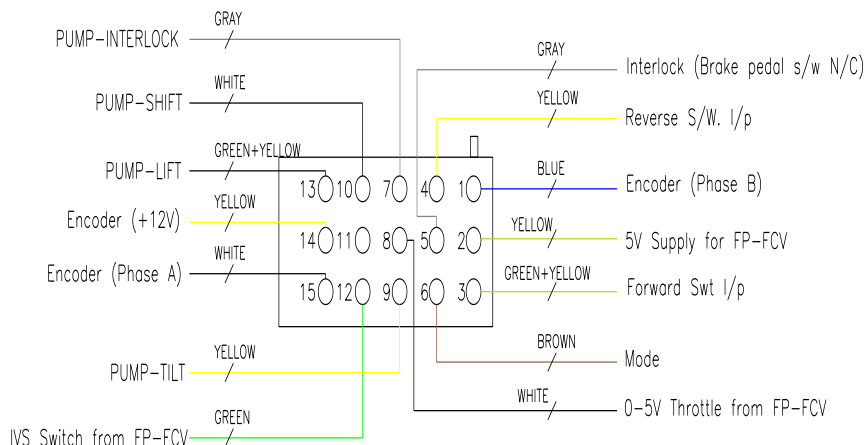


Sr. No.	Part No	Item Description	Qty	Unit
1	MH365925401	Traction controller -1236-5401AC,36-48V, 450A		
2	MH799208123	Traction controller -1236-5401AC,36-48V, 450A + Pump Controller -1234-5271AC,36-48V, 275A kit	1	No
3	MH378145271	Pump Controller -1234-5271AC,36-48V, 275A		
4	MH799208143	Main Wiring Harness AC+AC	1	No
5	MH799208063	Canopy Wiring Harness	1	No
6	MH61069525	Control panel wiring harness		
7	MH799208153	Power Cable Set AC+AC	1	Set
8	MH40468	FNR & Combination Switch (Honeywell)	1	No
9	MH512476400	12V DC Horn	1	No
10	MH18557	48V DC Reverse Horn	1	No
11	MH18555	48V DC LED Flasher	1	No
12	MH797016020	12V DC Head Light	2	No
13	MH10332	12V DC Signal & Parking light	2	No
14	MH42389/MH42388	Rear Indicator Light -RH/LH	2	
15	MH016051600	ON - OFF Key Switch - 48V, 5A	1	No
16	MH17840701-0003	SPY Glass (BDI)	1	No

POWER & CONTROL WIRING DIAGRAM FOR EVX20 MAX FULLY AC

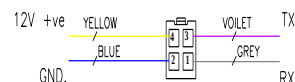




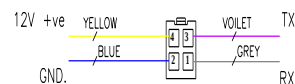


CN1 - 15 PIN POWERMATE CONN FEMALE

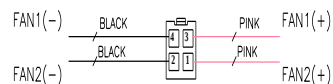
CN4 - 4 PIN MOLEX MALE



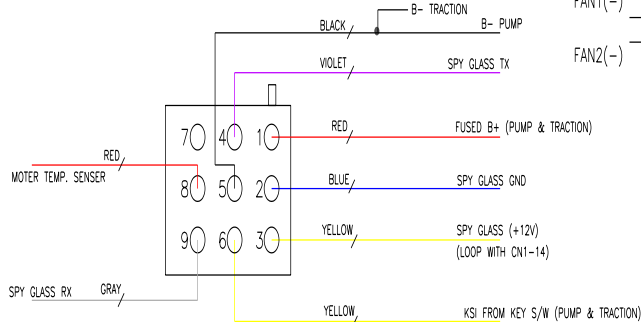
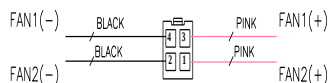
CN5 - 4 PIN MOLEX MALE



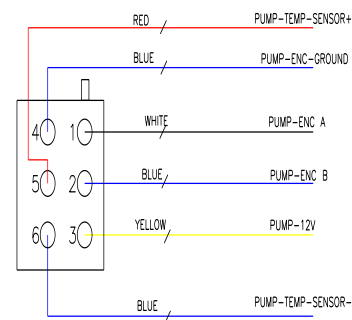
CN6 - 4 PIN MOLEX MALE & FEMALE



CN7 - 4 PIN MOLEX MALE & FEMALE



CN2 - 9 PIN POWERMATE CONN FEMALE



CN3 - 6 PIN POWERMATE CONN FEMALE

DIAGNOSTICS AND TROUBLESHOOTING

The 1236/38 controller detects a wide variety of faults or error conditions. Faults can be detected by the operating system or by the VCL code. This section describes the faults detected by the operating system. Faults detected by VCL code (faults 51–67 in Table 5) cannot be defined here as they will vary from application to application. Refer to the appropriate OEM documentation for information on these faults.

DIAGNOSTICS

Diagnostics information can be obtained in either of two ways: (1) by reading the display on a 1311 programmer or (2) by observing the fault codes issued by the Status LEDs. See Table 4 for a summary of LED display formats. The 1311 programmer will display all faults that are currently set as well as a history of the faults that have been set since the history log was last cleared. The 1311 displays the faults by name. The pair of LEDs built into the controller (one red, one yellow) produce flash codes displaying all the currently set faults in a repeating cycle. Each code consists of two digits. The red LED flashes once to indicate that the first digit of the code will follow; the yellow LED then flashes the appropriate number of times for the first digit. The red LED flashes twice to indicate that the second digit of the code will follow; the yellow LED flashes the appropriate number of times for the second digit.

Example: Battery Under voltage (code 23).

In the Fault menu of the 1311 programmer, the words

Example: Battery Undervoltage (code 23).

In the Fault menu of the 1311 programmer, the words **Undervoltage Cutback** will be displayed; the real-time battery voltage is displayed in the Monitor menu (“Keyswitch Voltage”).

The controller’s two LEDs will display this repeating pattern:

RED	YELLOW	RED	YELLOW
*	* *	* *	* * *
(first digit)	(2)	(second digit)	(3)

The numerical codes used by the yellow LED are listed in the troubleshooting chart (Table 5), which also lists possible fault causes and describes the conditions that set and clear each fault.

Summary of LED display formats

The two LEDs have four different display modes, indicating the type of information they are providing.

Table 4 TYPES OF LED DISPLAY	
DISPLAY	STATUS
Neither LED illuminated	Controller is not powered on, has a dead battery, or is severely damaged.
Yellow LED flashing	Controller is operating normally.
Yellow and red LEDs both on solid	Controller is in Flash program mode.
Red LED on solid	Watchdog failure. Cycle KSI to restart.
Red LED and yellow LED flashing alternately	Controller has detected a fault. 2-digit code flashed by yellow LED identifies the specific fault; one or two flashes by red LED indicate whether first or second code digit will follow.

TROUBLESHOOTING – 1236

The troubleshooting chart, Table 5, provides the following information on all the controller faults:

- fault code
- fault name as displayed on the programmer's LCD
- the effect of the fault
- possible causes of the fault
- fault set conditions
- fault clear conditions.

Whenever a fault is encountered and no wiring or vehicle fault can be found, shut off KSI and turn it back on to see if the fault clears. If not, shut off KSI and remove the 35-pin connector. Check the connector for corrosion or damage, clean it if necessary, and re-insert it.

Table 5 TROUBLESHOOTING CHART

CODE	PROGRAMMER LCD DISPLAY EFFECT OF FAULT	POSSIBLE CAUSE	SET/CLEAR CONDITIONS
12	Controller Overcurrent <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake.</i>	1. External short of phase U,V, or W motor connections. 2. Motor parameters are mis-tuned. 3. Controller defective.	<i>Set:</i> Phase current exceeded the current measurement limit. <i>Clear:</i> Cycle KSI.
13	Current Sensor Fault <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake.</i>	1. Leakage to vehicle frame from phase U, V, or W (short in motor stator). 2. Controller defective.	<i>Set:</i> Controller current sensors have invalid offset reading. <i>Clear:</i> Cycle KSI.
14	Precharge Failed <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake.</i>	1. External load on capacitor bank (B+ connection stud) that prevents the capacitor bank from charging. 2. See 1311 menu Monitor» Battery: Capacitor Voltage.	<i>Set:</i> Precharge failed to charge the capacitor bank to the KSI voltage. <i>Clear:</i> Cycle Interlock input or use VCL function <i>Precharge()</i> .
15	Controller Severe Undertemp <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake; ShutdownThrottle; FullBrake.</i>	1. Controller is operating in an extreme environment. 2. See 1311 menu Monitor» Controller: Temperature.	<i>Set:</i> Heatsink temperature below -40°C. <i>Clear:</i> Bring heatsink temperature above -40°C, and cycle interlock or KSI.
16	Controller Severe Overtemp <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake; ShutdownThrottle; FullBrake.</i>	1. Controller is operating in an extreme environment. 2. Excessive load on vehicle. 3. Improper mounting of controller. 4. See 1311 menu Monitor» Controller: Temperature.	<i>Set:</i> Heatsink temperature above +95°C. <i>Clear:</i> Bring heatsink temperature below +95°C, and cycle interlock or KSI.
17	Severe Undervoltage <i>Reduced drive torque.</i>	1. Battery Menu parameters are misadjusted. 2. Non-controller system drain on battery. 3. Battery resistance too high. 4. Battery disconnected while driving. 5. See 1311 menu Monitor» Battery: Capacitor Voltage. 6. Blown B+ fuse or main contactor did not close.	<i>Set:</i> Capacitor bank voltage dropped below the Severe Battery Undervoltage limit (see page 49) with FET bridge enabled. <i>Clear:</i> Bring capacitor voltage above Severe Battery Undervoltage limit.
18	Severe Overvoltage <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake; ShutdownThrottle; FullBrake.</i>	1. Battery Menu parameters are misadjusted. 2. Battery resistance too high for given regen current. 3. Battery disconnected while regen braking. 4. See 1311 menu Monitor» Battery: Capacitor Voltage.	<i>Set:</i> Capacitor bank voltage exceeded the Severe Battery Overvoltage limit (see page 49) with FET bridge enabled. <i>Clear:</i> Bring capacitor voltage below Severe Battery Overvoltage limit, and then cycle KSI.
21	Controller Undertemp Cutback <i>None, unless a fault action is programmed in VCL.</i>	1. Controller is performance-limited at this temperature. 2. Controller is operating in an extreme environment. 3. See 1311 menu Monitor» Controller: Temperature.	<i>Set:</i> Heatsink temperature dropped below -25°C. <i>Clear:</i> Bring heatsink temperature above -25°C.

Table 5 TROUBLESHOOTING CHART, continued

CODE	PROGRAMMER LCD DISPLAY EFFECT OF FAULT	POSSIBLE CAUSE	SET/CLEAR CONDITIONS
22	Controller Overtemp Cutback <i>Reduced drive and brake torque.</i>	1. Controller is performance-limited at this temperature. 2. Controller is operating in an extreme environment. 3. Excessive load on vehicle. 4. Improper mounting of controller. 5. See 1311 menu Monitor » Controller: Temperature.	<i>Set:</i> Heatsink temperature exceeded 85°C. <i>Clear:</i> Bring heatsink temperature below 85°C.
23	Undervoltage Cutback <i>Reduced drive torque.</i>	1. Normal operation. Fault shows that the batteries need recharging. Controller is performance limited at this voltage. 2. Battery parameters are misadjusted. 3. Non-controller system drain on battery. 4. Battery resistance too high. 5. Battery disconnected while driving. 6. See 1311 menu Monitor » Battery: Capacitor Voltage. 7. Blown B+ fuse or main contactor did not close.	<i>Set:</i> Capacitor bank voltage dropped below the Battery Undervoltage limit (see p. 49) with the FET bridge enabled. <i>Clear:</i> Bring capacitor voltage above the Undervoltage limit.
24	Overvoltage Cutback <i>Reduced brake torque.</i>	1. Normal operation. Fault shows that regen braking currents elevated the battery voltage during regen braking. Controller is performance limited at this voltage. 2. Battery parameters are misadjusted. 3. Battery resistance too high for given regen current. 4. Battery disconnected while regen braking. 5. See 1311 menu Monitor » Battery: Capacitor Voltage.	<i>Set:</i> Capacitor bank voltage exceeded the Battery Overvoltage limit (see page 49) with the FET bridge enabled. <i>Clear:</i> Bring capacitor voltage below the Overvoltage limit.
25	+5V Supply Failure <i>None, unless a fault action is programmed in VCL.</i>	1. External load impedance on the +5V supply (pin 26) is too low. 2. See 1311 menu Monitor » outputs: 5 Volts and Ext Supply Current.	<i>Set:</i> +5V supply (pin 26) outside the +5V±10% range. <i>Clear:</i> Bring voltage within range.
26	Digital Out 6 Overcurrent <i>Digital Output 6 driver will not turn on.</i>	1. External load impedance on Digital Output 6 driver (pin 19) is too low.	<i>Set:</i> Digital Output 6 (pin 19) current exceeded 15 mA. <i>Clear:</i> Remedy the overcurrent cause and use the VCL function <i>Set_DigOut()</i> to turn the driver on again.
27	Digital Out 7 Overcurrent <i>Digital Output 7 driver will not turn on.</i>	1. External load impedance on Digital Output 7 driver (pin 20) is too low.	<i>Set:</i> Digital Output 7 (pin 20) current exceeded 15 mA. <i>Clear:</i> Remedy the overcurrent cause and use the VCL function <i>Set_DigOut()</i> to turn the driver on again.

Table 5 TROUBLESHOOTING CHART, continued

CODE	PROGRAMMER LCD DISPLAY EFFECT OF FAULT	POSSIBLE CAUSE	SET/CLEAR CONDITIONS
28	Motor Temp Hot Cutback <i>Reduced drive torque.</i>	1. Motor temperature is at or above the programmed Temperature Hot setting, and the requested current is being cut back. 2. Motor Temperature Control Menu parameters are mis-tuned. 3. See 1311 menus Monitor » Motor: Temperature and Monitor » Inputs: Analog2. 4. If the application doesn't use a motor thermistor, Temp Compensation and Temp Cutback should be programmed Off.	<i>Set:</i> Motor temperature is at or above the Temperature Hot parameter setting. <i>Clear:</i> Bring the motor temperature within range.
29	Motor Temp Sensor Fault <i>Max.Speed reduced (LOS, Limited Operating Strategy) and motor temperature cutback is disabled.</i>	1. Motor thermistor is not connected properly. 2. If the application doesn't use a motor thermistor, Temp Compensation and Temp Cutback should be programmed Off. 3. See 1311 menus Monitor » Motor: Temperature and Monitor » Inputs: Analog2.	<i>Set:</i> Motor thermistor input (pin 8) is at the voltage rail (0 or 10V). <i>Clear:</i> Bring the motor thermistor input voltage within range.
31	Coil1 Driver Open/Short <i>ShutdownDriver1.</i>	1. Open or short on driver load. 2. Dirty connector pins. 3. Bad crimps or faulty wiring.	<i>Set:</i> Driver 1 (pin 6) is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver.
31	Main Open/Short <i>ShutdownDriver1; ShutdownMotor; ShutdownEMBrake.</i>	1. Open or short on driver load. 2. Dirty connector pins. 3. Bad crimps or faulty wiring.	<i>Set:</i> Main contactor driver (pin 6) is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver.
32	Coil2 Driver Open/Short <i>ShutdownDriver2.</i>	1. Open or short on driver load. 2. Dirty connector pins. 3. Bad crimps or faulty wiring.	<i>Set:</i> Driver 2 (pin 5) is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver.
32	EM Brake Open/Short <i>ShutdownDriver2; ShutdownThrottle; FullBrake.</i>	1. Open or short on driver load. 2. Dirty connector pins. 3. Bad crimps or faulty wiring.	<i>Set:</i> Electromagnetic brake driver (pin 5) is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver.
33	Coil3 Driver Open/Short <i>ShutdownDriver3.</i>	1. Open or short on driver load. 2. Dirty connector pins. 3. Bad crimps or faulty wiring.	<i>Set:</i> Driver 3 (pin 4) is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver.
34	Coil4 Driver Open/Short <i>ShutdownDriver4.</i>	1. Open or short on driver load. 2. Dirty connector pins. 3. Bad crimps or faulty wiring.	<i>Set:</i> Driver 4 (pin 3) is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver.
35	PD Open/Short <i>ShutdownPD.</i>	1. Open or short on driver load. 2. Dirty connector pins. 3. Bad crimps or faulty wiring.	<i>Set:</i> Proportional driver (pin 2) is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver.
36	Encoder Fault <i>Control Mode changed to LOS (Limited Operating Strategy).</i>	1. Motor encoder failure. 2. Bad crimps or faulty wiring. 3. See 1311 menu Monitor » Motor: Motor RPM.	<i>Set:</i> Motor encoder phase failure detected. <i>Clear:</i> Cycle KSI.

Table 5 TROUBLESHOOTING CHART, continued

CODE	PROGRAMMER LCD DISPLAY EFFECT OF FAULT	POSSIBLE CAUSE	SET/CLEAR CONDITIONS
37	Motor Open <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake.</i>	1. Motor phase is open. 2. Bad crimps or faulty wiring. 3. Bad crimps or faulty wiring.	<i>Set:</i> Motor phase U, V, or W detected open. <i>Clear:</i> Cycle KSI.
38	Main Contactor Welded <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake.</i>	1. Main contactor tips are welded closed. 2. Motor phase U is disconnected or open. 3. An alternate voltage path (such as an external precharge resistor) is providing a current to the capacitor bank (B+ connection stud).	<i>Set:</i> Just prior to the main contactor closing, the capacitor bank voltage (B+ connection stud) was loaded for a short time and the voltage did not discharge. <i>Clear:</i> Cycle KSI.
39	Main Contactor Did Not Close <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake.</i>	1. Main contactor did not close. 2. Main contactor tips are oxidized, burned, or not making good contact. 3. External load on capacitor bank (B+ connection stud) that prevents capacitor bank from charging. 4. Blown B+ fuse.	<i>Set:</i> With the main contactor commanded closed, the capacitor bank voltage (B+ connection stud) did not charge to B+. <i>Clear:</i> Cycle KSI.
41	Throttle Wiper High <i>ShutdownThrottle.</i>	1. Throttle pot wiper voltage too high. 2. See 1311 menu Monitor » Inputs: Throttle Pot.	<i>Set:</i> Throttle pot wiper (pin 16) voltage is higher than the high fault threshold (can be changed with the VCL function <i>Setup_Pot_Faults()</i>). <i>Clear:</i> Bring throttle pot wiper voltage below the fault threshold.
42	Throttle Wiper Low <i>ShutdownThrottle.</i>	1. Throttle pot wiper voltage too low. 2. See 1311 menu Monitor » Inputs: Throttle Pot.	<i>Set:</i> Throttle pot wiper (pin 16) voltage is lower than the low fault threshold (can be changed with the VCL function <i>Setup_Pot_Faults()</i>). <i>Clear:</i> Bring throttle pot wiper voltage above the fault threshold.
43	Brake Wiper High <i>FullBrake.</i>	1. Brake pot wiper voltage too high. 2. See 1311 menu Monitor » Inputs: Brake Pot.	<i>Set:</i> Brake pot wiper (pin 17) voltage is higher than the high fault threshold (can be changed with the VCL function <i>Setup_Pot_Faults()</i>). <i>Clear:</i> Bring brake pot wiper voltage below the fault threshold.
44	Brake Wiper Low <i>FullBrake.</i>	1. Brake pot wiper voltage too low. 2. See 1311 menu Monitor » Inputs: Brake Pot.	<i>Set:</i> Brake pot wiper (pin 17) voltage is lower than the low fault threshold (can be changed with the VCL function <i>Setup_Pot_Faults()</i>). <i>Clear:</i> Bring brake pot wiper voltage above the fault threshold.
45	Pot Low Overcurrent <i>ShutdownThrottle; FullBrake.</i>	1. Combined pot resistance connected to pot low is too low. 2. See 1311 menu Monitor » Outputs: Pot Low.	<i>Set:</i> Pot low (pin 18) current exceeds 10mA. <i>Clear:</i> Clear pot low overcurrent condition and cycle KSI.

Table 5 TROUBLESHOOTING CHART, continued

CODE	PROGRAMMER LCD DISPLAY EFFECT OF FAULT	POSSIBLE CAUSE	SET/CLEAR CONDITIONS
46	EEPROM Failure <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake; ShutdownThrottle; ShutdownInterlock; ShutdownDriver1; ShutdownDriver2; ShutdownDriver3; ShutdownDriver4; ShutdownPD; FullBrake.</i>	1. Failure to write to EEPROM memory. This can be caused by EEPROM memory writes initiated by VCL, by the CAN bus, by adjusting parameters with the 1311, or by loading new software into the controller.	<i>Set:</i> Controller operating system tried to write to EEPROM memory and failed. <i>Clear:</i> Download the correct software (OS) and matching parameter default settings into the controller and cycle KSI.
47	HPD/Sequencing Fault <i>ShutdownThrottle.</i>	1. KSI, interlock, direction, and throttle inputs applied in incorrect sequence. 2. Faulty wiring, crimps, or switches at KSI, interlock, direction, or throttle inputs. 3. See 1311 menu Monitor»Inputs.	<i>Set:</i> HPD (High Pedal Disable) or sequencing fault caused by incorrect sequence of KSI, interlock, direction, and throttle inputs. <i>Clear:</i> Reapply inputs in correct sequence.
49	Parameter Change Fault <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake.</i>	1. This is a safety fault caused by a change in certain 1311 parameter settings so that the vehicle will not operate until KSI is cycled. For example, if a user changes the Throttle Type this fault will appear and require cycling KSI before the vehicle can operate.	<i>Set:</i> Adjustment of a parameter setting that requires cycling of KSI. <i>Clear:</i> Cycle KSI.
51–67	OEM Faults (See OEM documentation.)	1. These faults can be defined by the OEM and are implemented in the application-specific VCL code. See OEM documentation.	<i>Set:</i> See OEM documentation. <i>Clear:</i> See OEM documentation.
68	VCL Runtime Error <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake; ShutdownThrottle; ShutdownInterlock; ShutdownDriver1; ShutdownDriver2; ShutdownDriver3; ShutdownDriver4; ShutdownPD; FullBrake.</i>	1. VCL code encountered a runtime VCL error. 2. See 1311 menu Monitor»Controller: VCL Error Module and VCL Error. This error can then be compared to the runtime VCL module ID and error code definitions found in the specific OS system information file.	<i>Set:</i> Runtime VCL code error condition. <i>Clear:</i> Edit VCL application software to fix this error condition; flash the new compiled software and matching parameter defaults; cycle KSI.
69	External Supply Out of Range <i>None, unless a fault action is programmed in VCL.</i>	1. External load on the 5V and 12V supplies draws either too much or too little current. 2. Fault Checking Menu parameters Ext Supply Max and Ext Supply Min are mis-tuned. 3. See 1311 menu Monitor»Outputs: Ext Supply Current.	<i>Set:</i> The external supply current (combined current used by the 5V supply [pin 26] and 12V supply [pin 25]) is either greater than the upper current threshold or lower than the lower current threshold. The two thresholds are defined by the Ext Supply Max and Ext Supply Min parameter settings (page 45). <i>Clear:</i> Bring the external supply current within range.

Table 5 TROUBLESHOOTING CHART, continued

CODE	PROGRAMMER LCD DISPLAY EFFECT OF FAULT	POSSIBLE CAUSE	SET/CLEAR CONDITIONS
71	OS General <i>ShutdownMainContactor; ShutdownMotor; ShutdownEMBrake; ShutdownThrottle; ShutdownInterlock; ShutdownDriver1; ShutdownDriver2; ShutdownDriver3; ShutdownDriver4; ShutdownPD; FullBrake.</i>	1. Internal controller fault.	<i>Set:</i> Internal controller fault detected. <i>Clear:</i> Cycle KSI.
72	PDO Timeout <i>ShutdownInterlock; CAN NMT State set to Pre-operational.</i>	1. Time between CAN PDO messages received exceeded the PDO Timeout Period.	<i>Set:</i> Time between CAN PDO messages received exceeded the PDO Timeout Period. <i>Clear:</i> Cycle KSI.
73	Stall Detect <i>Control Mode changed to LOS (Limited Operating Strategy).</i>	1. Stalled motor. 2. Motor encoder failure. 3. Bad crimps or faulty wiring. 4. Problems with power supply for the motor encoder. 5. See 1311 menu Monitor» Motor: Motor RPM.	<i>Set:</i> No motor encoder movement detected. <i>Clear:</i> Either cycle KSI, or detect valid motor encoder signals while operating in LOS mode and return Throttle Command = 0 and Motor RPM = 0.

MAINTENANCE

There are no user serviceable parts in the Curtis 1236/38 controller. **No attempt**

should be made to open, repair, or otherwise modify the controller. Doing so may damage the controller and will void the warranty. It is recommended that the controller and connections be kept clean and dry and that the controller's diagnostics history file be checked and cleared periodically.

CLEANING

Periodically cleaning the controller exterior will help protect it against corrosion and possible electrical control problems created by dirt, grime, and chemicals that are part of the operating environment and that normally exist in battery

powered systems.

When working around any battery powered system, proper safety precautions should be taken. These include, but are not limited to: proper training, wearing eye protection, and avoiding loose clothing and jewelry.

Use the following cleaning procedure for routine maintenance. Never use a high pressure washer to clean the controller.

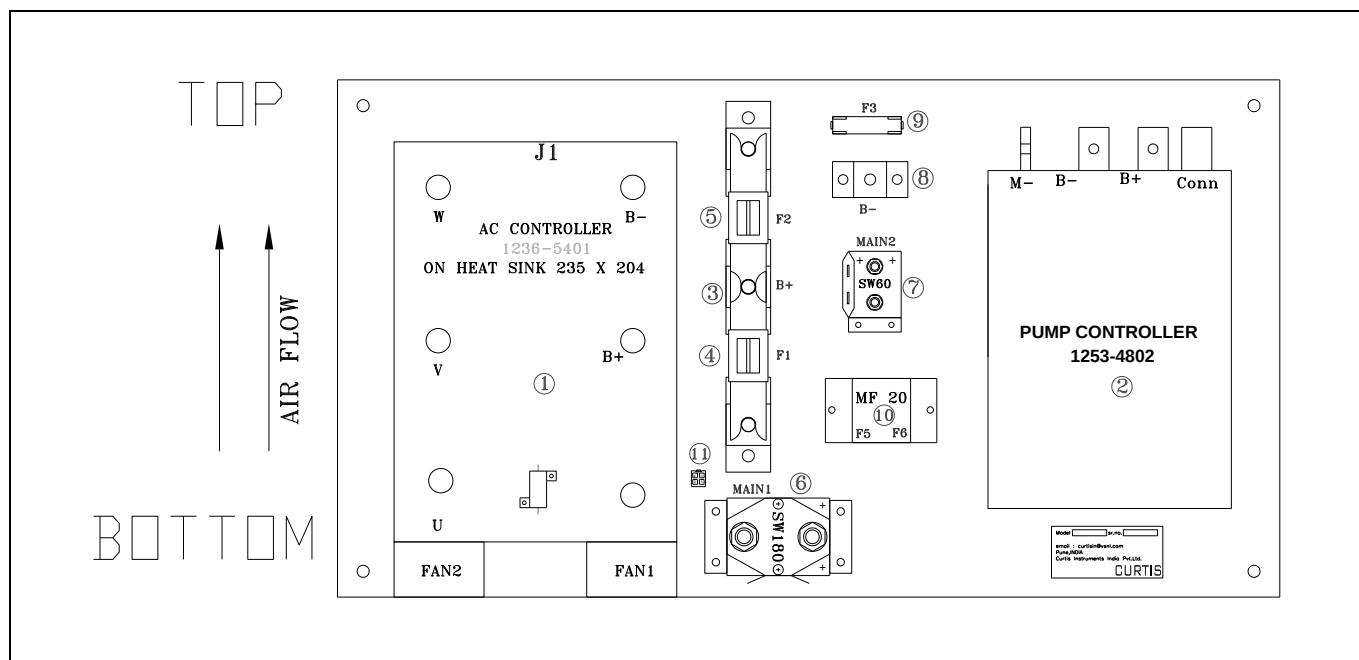
1. Remove power by disconnecting the battery.
2. Discharge the capacitors in the controller by connecting a load (such as a contactor coil) across the controller's **B+** and **B-** terminals.
3. Remove any dirt or corrosion from the power and signal connector areas. The controller should be wiped clean with a moist rag. Dry it before reconnecting the battery.
4. Make sure the connections are tight. Refer to Section 2, page 6, for maximum tightening torque specifications for the battery and motor connections.

DIAGNOSTIC HISTORY

The 1311 programmer can be used to access the controller's diagnostic history file. The programmer will read out all the faults the controller has experienced since the last time the diagnostic history file was cleared. Faults such as contactor faults may be the result of loose wires; contactor wiring should be carefully checked. Faults such as overtemperature may be caused by operator habits or by overloading.

After a problem has been diagnosed and corrected, it is a good idea to clear the diagnostic history file. This allows the controller to accumulate a new file of faults. By checking the new diagnostic history file at a later date, you can readily determine whether the problem was indeed fixed.

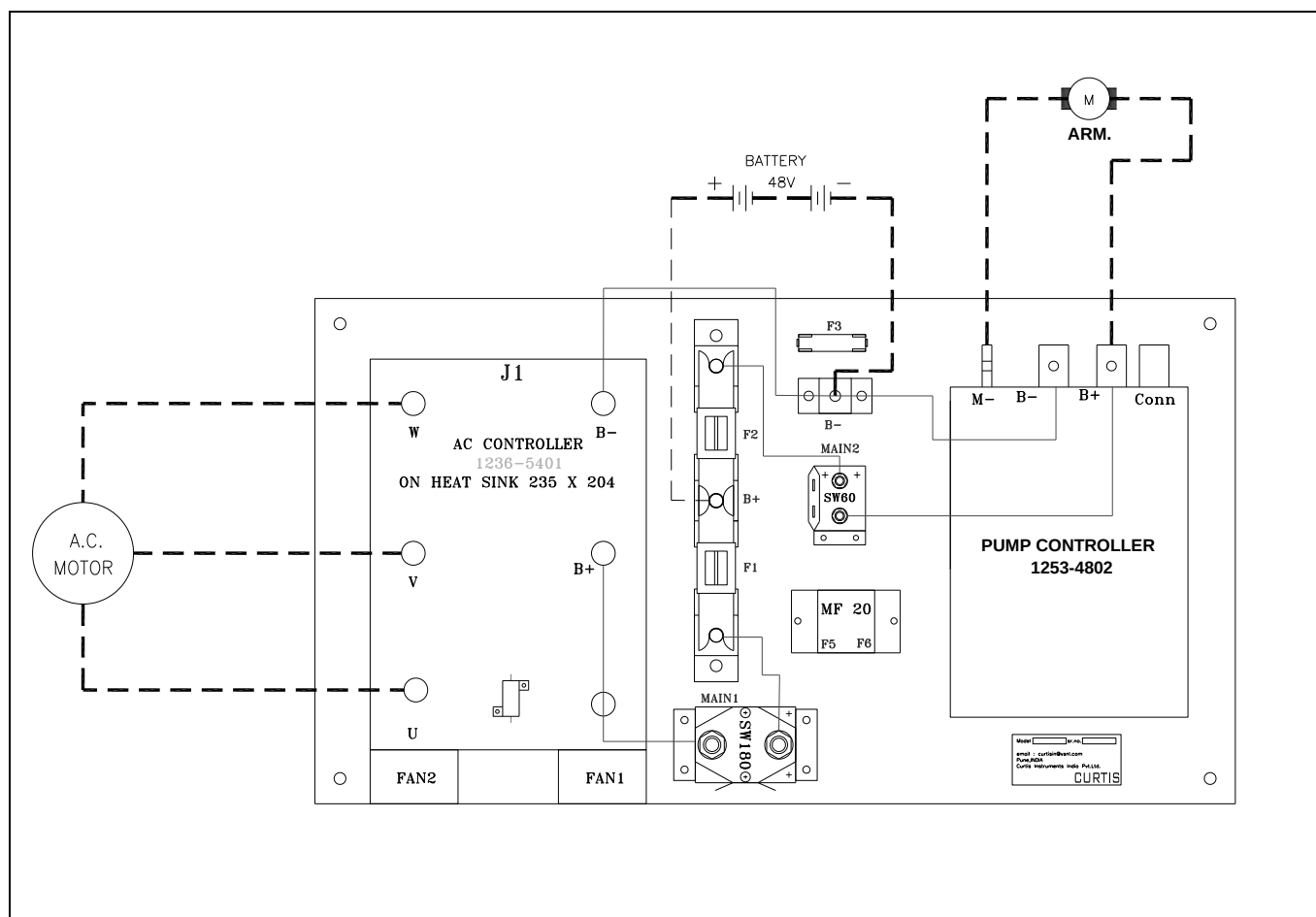
COMPONENT LAYOUT EVX20MAX AC/DC



ELECTRICAL SPARES EVX20MAX AC/DC

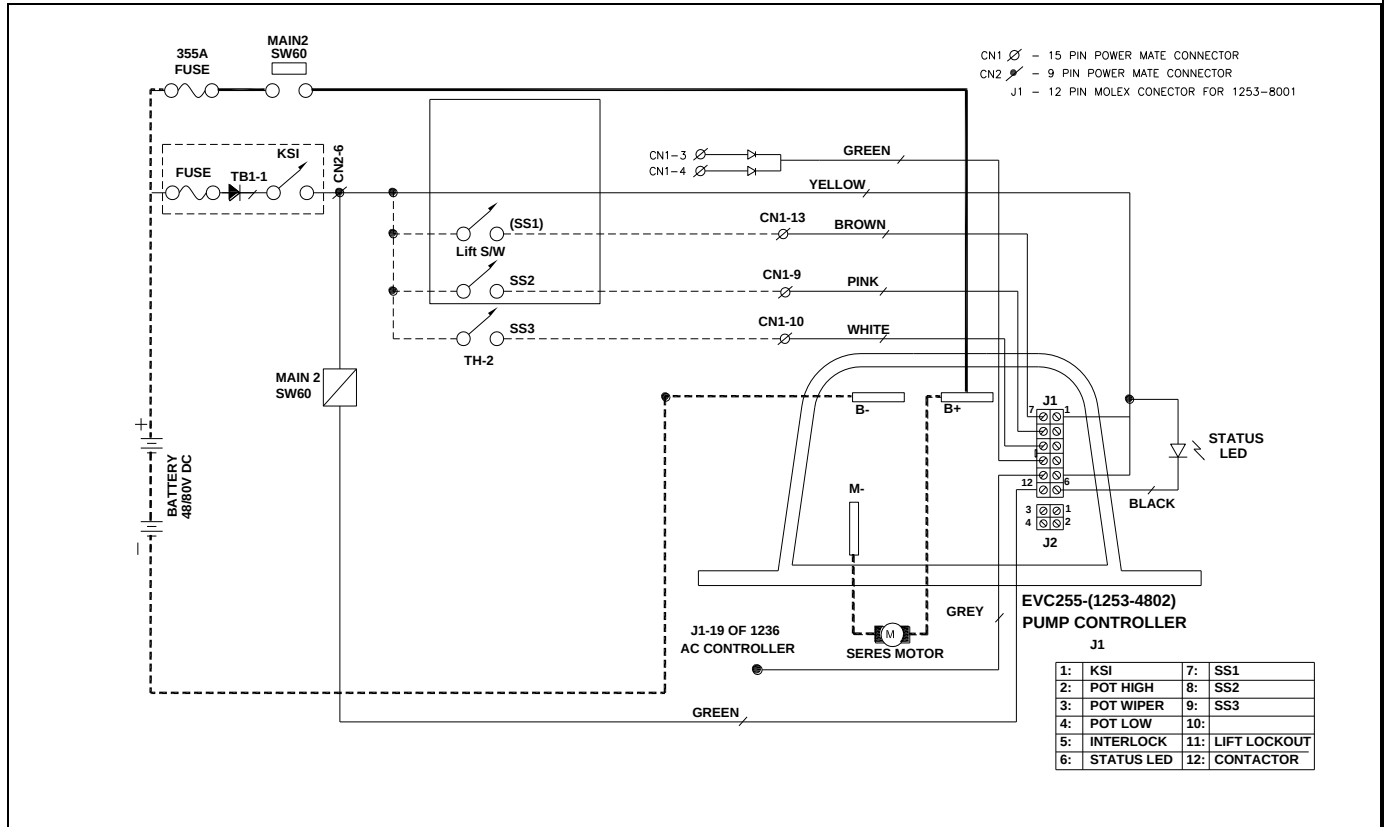
Sr. No.	Item Description	Part no	Qty
1	Traction Controller - 1236-5401 AC, 36-48V, 450A	MH365925401	1No
2	Pump Controller - 1253 -4802 DC, 48V, 600A	MH17724701	1No
3	Battery Charger	MH17264	1No
4	MAIN1 Fuse 250A	MH29221	1No
5	MAIN2 Fuse 355A	MH14290A0002	1No
6	Pump Contactor SU80, 48V	MH13223696	1No
7	MAIN2 Contactor SU60, 48V	MH13222745	1No
8	Control Panel harness	MH61069395	1No
9	Accelerator Assy with Potentiometer	18140800-0003	1No
10	Battery connector	MH17840701-003	2No
11	Cooling fan for controller(24V)	Mh14290A0001	2No

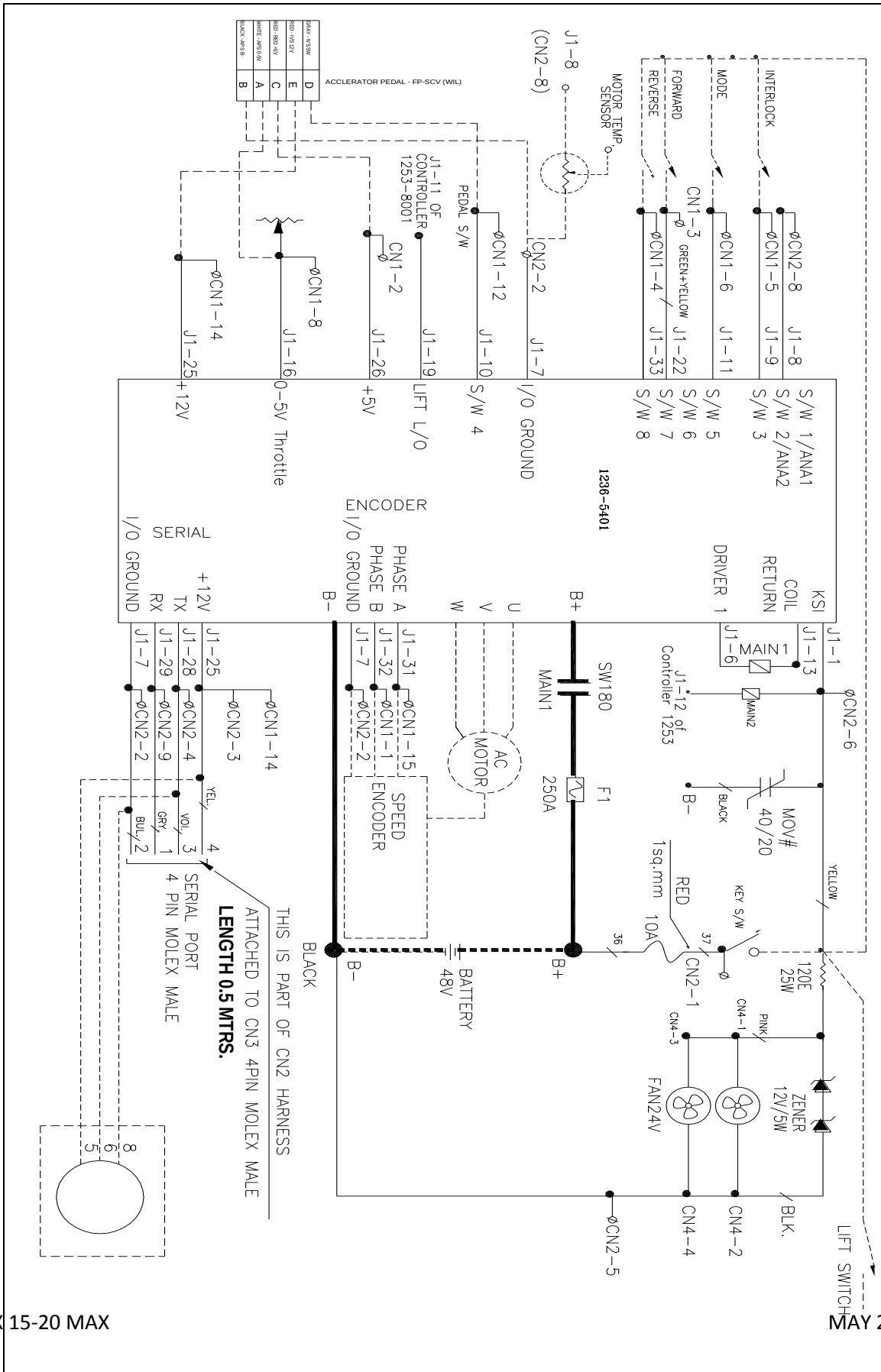
POWER WIRING DIAGRAM EVX20MAX AC /DC

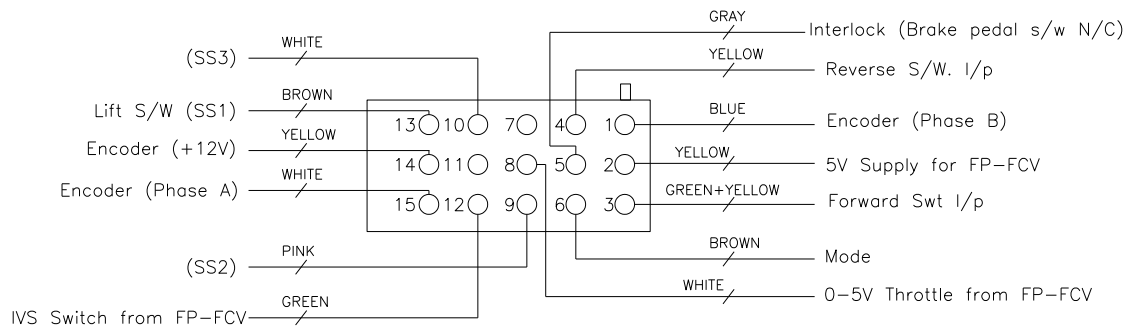


Sr. No.	Part No	Item Description	Qty	Unit
1	MH799208013	Traction Controller - 1236-5401 AC, 36-48V, 450A + Pump Controller - 1253 -4802 DC, 48V, 600A + Harness kit	1	No
2	MH799208053	Main Wiring Harness(AC/DC)	1	No
3	MH799208073	Canopy Wiring Harness	1	No
4	**MH799208073	Power Cable Set	1	No
5	MH40468	FNR & Combination Switch (Honeywell)	1	No
6	MH512476400	12V DC Horn	1	No
7	MH18557	48V DC Reverse Horn	1	No
8	MH18555	48V DC LED Flasher	1	No
9	*MH797016020	12V DC Head Light	2	No
10	MH80124801123	Combination tail light	1	No
11	MH17840701-0003	Spy Glass(BDI)		No
12	MH016051600	ON - OFF Key Switch - 48V, 5A	1	No
9	MH42324	Indicator light front	2	
10	MH08995	STOP LIGHT S/W	1	
11	MH797017393	BATTERY CONECTOR HANDLE	1	
12	MH41005	48V 440AH	1181	

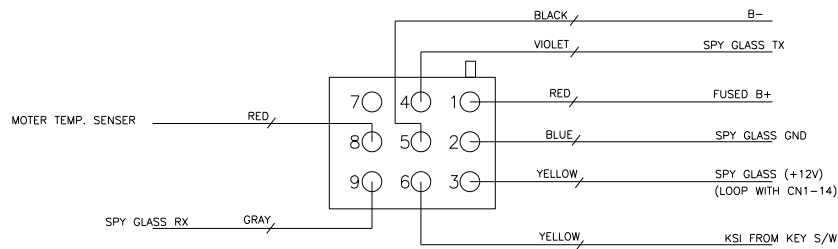
POWER & CONTROL WIRING DIAGRAM FOR DC PUMP CONTROLLER



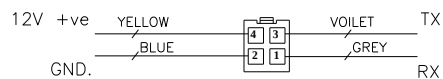




CN1 - 15 PIN POWERMATE CONN



CN2 - 9 PIN POWERMATE CONN



CN3 - 4 PIN MOLEX MALE

TITLE - CONNECTOR DETAILS FOR
BPA - PNL-081-14AC
CPA - PNL-081-18

DIAGNOSTICS AND TROUBLESHOOTING – 1253 – pump controller

The 1253 controller provides diagnostics information to assist technicians in troubleshooting pump system problems. The diagnostics information can be obtained by observing the appropriate display on the handheld programmer or the fault codes issued by the optional Status LED. Refer to the troubleshooting chart (Table 5) for suggestions covering a wide range of possible faults.

PROGRAMMER DIAGNOSTICS

The programmer presents complete diagnostic information in plain language. Faults are displayed in the Faults Menu (see column 2 in the troubleshooting chart), and the status of the controller inputs/outputs is displayed in the Monitor Menu.

Accessing the Fault History Menu provides a list of the faults that have occurred since the fault history file was last cleared. Checking (and clearing)

the fault history file is recommended each time the vehicle is brought in for maintenance.

The following 4-step process is recommended for diagnosing and troubleshooting an inoperative pump system: (1) visually inspect the vehicle for obvious problems; (2) diagnose the problem, using the programmer; (3) test the circuitry with the programmer; and (4) correct the problem. Repeat the last three steps as necessary until the pump system is operational.

Example: A vehicle that cannot perform the operation requested by Speed Select 2 is brought in for repair.

STEP 1: Examine the vehicle and its wiring for any obvious problems, such as broken wires or loose connections.

STEP 2: Connect the programmer, select the Faults Menu, and read the displayed fault information. In this example, the display shows “No Known Faults,” indicating that the controller has not detected anything out of the norm.

STEP 3: Select the Monitor Menu, and observe the status of the SS2 input. In this example, the display shows that the switch does not close when SS2 is selected, which means the problem is either in the SS2 switch or the switch wiring.

STEP 4: Check or replace the SS2 switch and wiring and repeat the test. If the programmer shows the SS2 switch closing and the system now operates normally, the problem has been corrected.

Table 5 TROUBLESHOOTING CHART

LED CODE	PROGRAMMER LCD DISPLAY	EXPLANATION	POSSIBLE CAUSE
1,1	EEPROM FAULT	EEPROM fault. <i>Note: Usually can be cleared by modifying any parameter value in the Program Menu.</i>	1. EEPROM data lost or damaged. 2. EEPROM checksum error.
1,2	HW FAILSAFE	Self-test or watchdog fault.	1. MOSFET shorted. 2. Controller defective.
1,3	MOTOR SHORTED	Motor shorted.	1. Motor is shorted.
2,1	UNDERVOLTAGE CUTOFF	Undervoltage cutoff.	1. Battery voltage < LOVOLT CUTOFF setting.
2,2	LIFT LOCKOUT	Lift operation locked out due to undervoltage.	1. Controller received appropriate lift lock-out signal. 2. Inappropriate lift lockout signal: SS LOCKOUT parameter not set correctly.
2,3	SEQUENCE ERROR	Startup lockout.	1. Improper sequence of throttle or SS and KSI or KSI plus interlock. 2. STARTUP LOCKOUT parameter not set correctly. 3. Misadjusted throttle.
2,4	THROTTLE FAULT	Wiper signal out of range (pot low fault).	1. Throttle input wire open or shorted. 2. Throttle defective. 3. THROTTLE TYPE parameter not set correctly.
3,1	CONT DRVR OC	Main contactor coil overcurrent.	1. Main contactor coil shorted. 2. Controller defective.
3,2	MAIN CONT WELDED	Main contactor welded.	1. Main contactor stuck closed. 2. CONT CNTRL parameter not set correctly. 3. Main contactor driver shorted.
3,3	PRECHARGE FAULT	Precharge fault.	1. Precharge circuit failure. 2. External short or leakage between B+ and B-.
3,4	MAIN CONT DNC	Main contactor did not close.	1. Main contactor coil connection loose. 2. Main contactor did not close. 3. CONT CNTRL parameter not set correctly.
4,1	LOW BATTERY VOLTAGE	Low battery voltage.	1. Battery voltage < undervoltage cutback threshold. 2. Corroded battery terminal. 3. Loose battery or controller terminal.
4,2	OVERVOLTAGE	Overvoltage.	1. Battery voltage > overvoltage shutdown threshold. 2. Vehicle operating with charger attached.
4,3	THERMAL CUTBACK	Over-/undertemperature cutback.	1. Temperature > 85°C or < -25°C. 2. Excessive load on pump motor. 3. Improper mounting of controller 4. Operation in extreme environment. 5. Thermistor failure.

MAINTENANCE

There are no user serviceable parts in the Curtis 1253 controller. **No attempt should be made to open, repair, or otherwise modify the controller.** Doing so may damage the controller and will void the warranty. It is recommended that the controller be kept **clean and dry** that its fault history file be checked and cleared periodically.

CLEANING

Periodically cleaning the controller exterior will help protect it against corrosion and possible electrical control problems created by dirt, grime, and chemicals that are part of the operating environment and that normally exist in battery powered systems.

When working around any battery powered system, proper safety precautions should be taken. These include, but are not limited to: proper training, wearing eye protection, and avoiding loose clothing and jewelry. Use the following cleaning procedure for routine maintenance. Never use a high pressure washer to clean the controller.

1. Remove power by disconnecting the battery.
2. Discharge the capacitors in the controller by connecting a load (such as a contactor coil) across the controller's **B+** and **B-** terminals.
3. Remove any dirt or corrosion from the power and signal connector areas. The controller should be wiped clean with a moist rag. Dry it before reconnecting the battery.
4. Make sure the connections are tight.

FAULT HISTORY

The programmer can be used to access the controller's fault history file. The programmer will read out all the faults that the controller has experienced since the last time the fault history file was cleared. The faults may be intermittent faults, faults caused

by loose wires, or faults caused by operator errors. Faults such as contactor faults may be the result of loose wires; contactor wiring should be carefully checked. Faults such as startup lockout or overtemperature may be caused by operator habits or by overloading. After a problem has been diagnosed and corrected, it is a good idea to clear the diagnostic history file. This allows the controller to accumulate a new file of faults. By checking the new fault history file at a later date, you can readily determine whether the problem was indeed fixed.